

Project 2 Data 624

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1 Project 2

1.1 Importing the data

Recap located at the end of the section. Skip there to jump past the code/workflow.

```
library(tidyverse)
library(VIM)

test_data <- read.csv("https://raw.githubusercontent.com/samanthabarbaro/data624/refs/heads/main/testdata.csv")
training_data <- read.csv("https://raw.githubusercontent.com/samanthabarbaro/data624/refs/heads/main/trainingdata.csv")

glimpse(test_data)
```

```
## Rows: 267
## Columns: 33
## $ Brand.Code      <chr> "D", "A", "B", "B", "B", "B", "A", "B", "A", "D", "B~
## $ Carb.Volume     <dbl> 5.480000, 5.393333, 5.293333, 5.266667, 5.406667, 5.~
## $ Fill.Ounces     <dbl> 24.03333, 23.95333, 23.92000, 23.94000, 24.20000, 24~
## $ PC.Volume       <dbl> 0.2700000, 0.2266667, 0.3033333, 0.1860000, 0.160000~
## $ Carb.Pressure   <dbl> 65.4, 63.2, 66.4, 64.8, 69.4, 73.4, 65.2, 67.4, 66.8~
## $ Carb.Temp       <dbl> 134.6, 135.0, 140.4, 139.0, 142.2, 147.2, 134.6, 139~
## $ PSC             <dbl> 0.236, 0.042, 0.068, 0.004, 0.040, 0.078, 0.088, 0.0~
## $ PSC.Fill        <dbl> 0.40, 0.22, 0.10, 0.20, 0.30, 0.22, 0.14, 0.10, 0.48~
## $ PSC.CO2         <dbl> 0.04, 0.08, 0.02, 0.02, 0.06, NA, 0.00, 0.04, 0.04, ~
## $ Mnf.Flow        <dbl> -100, -100, -100, -100, -100, -100, -100, -100, -100~
## $ Carb.Pressure1  <dbl> 116.6, 118.8, 120.2, 124.8, 115.0, 118.6, 117.6, 121~
## $ Fill.Pressure   <dbl> 46.0, 46.2, 45.8, 40.0, 51.4, 46.4, 46.2, 40.0, 43.8~
## $ Hyd.Pressure1   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure2   <dbl> NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ Hyd.Pressure3   <dbl> NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ Hyd.Pressure4   <int> 96, 112, 98, 132, 94, 94, 108, 108, 110, 106, 98, 96~
## $ Filler.Level    <dbl> 129.4, 120.0, 119.4, 120.2, 116.0, 120.4, 119.6, 131~
## $ Filler.Speed    <int> 3986, 4012, 4010, NA, 4018, 4010, 4010, NA, 4010, 10~
## $ Temperature     <dbl> 66.0, 65.6, 65.6, 74.4, 66.4, 66.6, 66.8, NA, 65.8, ~
## $ Usage.cont      <dbl> 21.66, 17.60, 24.18, 18.12, 21.32, 18.00, 17.68, 12.~
## $ Carb.Flow       <int> 2950, 2916, 3056, 28, 3214, 3064, 3042, 1972, 2502, ~
## $ Density         <dbl> 0.88, 1.50, 0.90, 0.74, 0.88, 0.84, 1.48, 1.60, 1.52~
## $ MFR             <dbl> 727.6, 735.8, 734.8, NA, 752.0, 732.0, 729.8, NA, 74~
## $ Balling         <dbl> 1.398, 2.942, 1.448, 1.056, 1.398, 1.298, 2.894, 3.3~
## $ Pressure.Vacuum <dbl> -3.8, -4.4, -4.2, -4.0, -4.0, -3.8, -4.2, -4.4, -4.4~
## $ PH              <lg1> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ Oxygen.Filler   <dbl> 0.022, 0.030, 0.046, NA, 0.082, 0.064, 0.042, 0.096,~
## $ Bowl.Setpoint   <int> 130, 120, 120, 120, 120, 120, 120, 120, 120, 120, 12~
## $ Pressure.Setpoint <dbl> 45.2, 46.0, 46.0, 46.0, 50.0, 46.0, 46.0, 46.0, 46.0~
## $ Air.Pressurer   <dbl> 142.6, 147.2, 146.6, 146.4, 145.8, 146.0, 145.0, 146~
## $ Alch.Rel        <dbl> 6.56, 7.14, 6.52, 6.48, 6.50, 6.50, 7.18, 7.16, 7.14~
## $ Carb.Rel        <dbl> 5.34, 5.58, 5.34, 5.50, 5.38, 5.42, 5.46, 5.42, 5.44~
## $ Balling.Lvl     <dbl> 1.48, 3.04, 1.46, 1.48, 1.46, 1.44, 3.02, 3.00, 3.10~
```

```
glimpse(training_data)
```

```
## Rows: 2,571
## Columns: 33
## $ Brand.Code      <chr> "B", "A", "B", "A", "A", "A", "A", "B", "B", "B", "B~
## $ Carb.Volume     <dbl> 5.340000, 5.426667, 5.286667, 5.440000, 5.486667, 5.~
## $ Fill.Ounces     <dbl> 23.96667, 24.00667, 24.06000, 24.00667, 24.31333, 23~
## $ PC.Volume       <dbl> 0.2633333, 0.2386667, 0.2633333, 0.2933333, 0.111333~
## $ Carb.Pressure   <dbl> 68.2, 68.4, 70.8, 63.0, 67.2, 66.6, 64.2, 67.6, 64.2~
## $ Carb.Temp       <dbl> 141.2, 139.6, 144.8, 132.6, 136.8, 138.4, 136.8, 141~
## $ PSC             <dbl> 0.104, 0.124, 0.090, NA, 0.026, 0.090, 0.128, 0.154,~
## $ PSC.Fill        <dbl> 0.26, 0.22, 0.34, 0.42, 0.16, 0.24, 0.40, 0.34, 0.12~
## $ PSC.CO2         <dbl> 0.04, 0.04, 0.16, 0.04, 0.12, 0.04, 0.04, 0.04, 0.14~
## $ Mnf.Flow        <dbl> -100, -100, -100, -100, -100, -100, -100, -100, -100~
## $ Carb.Pressure1  <dbl> 118.8, 121.6, 120.2, 115.2, 118.4, 119.6, 122.2, 124~
## $ Fill.Pressure   <dbl> 46.0, 46.0, 46.0, 46.4, 45.8, 45.6, 51.8, 46.8, 46.0~
## $ Hyd.Pressure1   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure2   <dbl> NA, NA, NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure3   <dbl> NA, NA, NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Hyd.Pressure4   <int> 118, 106, 82, 92, 92, 116, 124, 132, 90, 108, 94, 86~
## $ Filler.Level    <dbl> 121.2, 118.6, 120.0, 117.8, 118.6, 120.2, 123.4, 118~
## $ Filler.Speed    <int> 4002, 3986, 4020, 4012, 4010, 4014, NA, 1004, 4014, ~
## $ Temperature     <dbl> 66.0, 67.6, 67.0, 65.6, 65.6, 66.2, 65.8, 65.2, 65.4~
## $ Usage.cont      <dbl> 16.18, 19.90, 17.76, 17.42, 17.68, 23.82, 20.74, 18.~
## $ Carb.Flow       <int> 2932, 3144, 2914, 3062, 3054, 2948, 30, 684, 2902, 3~
## $ Density         <dbl> 0.88, 0.92, 1.58, 1.54, 1.54, 1.52, 0.84, 0.84, 0.90~
## $ MFR             <dbl> 725.0, 726.8, 735.0, 730.6, 722.8, 738.8, NA, NA, 74~
## $ Balling         <dbl> 1.398, 1.498, 3.142, 3.042, 3.042, 2.992, 1.298, 1.2~
## $ Pressure.Vacuum <dbl> -4.0, -4.0, -3.8, -4.4, -4.4, -4.4, -4.4, -4.4, -4.4~
## $ PH              <dbl> 8.36, 8.26, 8.94, 8.24, 8.26, 8.32, 8.40, 8.38, 8.38~
## $ Oxygen.Filler   <dbl> 0.022, 0.026, 0.024, 0.030, 0.030, 0.024, 0.066, 0.0~
## $ Bowl.Setpoint   <int> 120, 120, 120, 120, 120, 120, 120, 120, 120, 120, 12~
## $ Pressure.Setpoint <dbl> 46.4, 46.8, 46.6, 46.0, 46.0, 46.0, 46.0, 46.0, 46.0~
## $ Air.Pressurer    <dbl> 142.6, 143.0, 142.0, 146.2, 146.2, 146.6, 146.2, 146~
## $ Alch.Rel        <dbl> 6.58, 6.56, 7.66, 7.14, 7.14, 7.16, 6.54, 6.52, 6.52~
## $ Carb.Rel        <dbl> 5.32, 5.30, 5.84, 5.42, 5.44, 5.44, 5.38, 5.34, 5.34~
## $ Balling.Lvl     <dbl> 1.48, 1.56, 3.28, 3.04, 3.04, 3.02, 1.44, 1.44, 1.44~
```

```
str(training_data)
```

```
## 'data.frame': 2571 obs. of 33 variables:
## $ Brand.Code : chr "B" "A" "B" "A" ...
## $ Carb.Volume : num 5.34 5.43 5.29 5.44 5.49 ...
## $ Fill.Ounces : num 24 24 24.1 24 24.3 ...
## $ PC.Volume : num 0.263 0.239 0.263 0.293 0.111 ...
## $ Carb.Pressure : num 68.2 68.4 70.8 63 67.2 66.6 64.2 67.6 64.2 72 ...
## $ Carb.Temp : num 141 140 145 133 137 ...
## $ PSC : num 0.104 0.124 0.09 NA 0.026 0.09 0.128 0.154 0.132 0.014 ...
## $ PSC.Fill : num 0.26 0.22 0.34 0.42 0.16 0.24 0.4 0.34 0.12 0.24 ...
## $ PSC.CO2 : num 0.04 0.04 0.16 0.04 0.12 0.04 0.04 0.04 0.14 0.06 ...
## $ Mnf.Flow : num -100 -100 -100 -100 -100 -100 -100 -100 -100 -100 ...
## $ Carb.Pressure1 : num 119 122 120 115 118 ...
```

```
## $ Fill.Pressure      : num  46 46 46 46.4 45.8 45.6 51.8 46.8 46 45.2 ...
## $ Hyd.Pressure1      : num   0 0 0 0 0 0 0 0 0 0 ...
## $ Hyd.Pressure2      : num  NA NA NA 0 0 0 0 0 0 0 ...
## $ Hyd.Pressure3      : num  NA NA NA 0 0 0 0 0 0 0 ...
## $ Hyd.Pressure4      : int  118 106 82 92 92 116 124 132 90 108 ...
## $ Filler.Level       : num  121 119 120 118 119 ...
## $ Filler.Speed       : int  4002 3986 4020 4012 4010 4014 NA 1004 4014 4028 ...
## $ Temperature       : num   66 67.6 67 65.6 65.6 66.2 65.8 65.2 65.4 66.6 ...
## $ Usage.cont         : num   16.2 19.9 17.8 17.4 17.7 ...
## $ Carb.Flow          : int  2932 3144 2914 3062 3054 2948 30 684 2902 3038 ...
## $ Density            : num   0.88 0.92 1.58 1.54 1.54 1.52 0.84 0.84 0.9 0.9 ...
## $ MFR                : num   725 727 735 731 723 ...
## $ Balling            : num   1.4 1.5 3.14 3.04 3.04 ...
## $ Pressure.Vacuum    : num   -4 -4 -3.8 -4.4 -4.4 -4.4 -4.4 -4.4 -4.4 -4.4 ...
## $ PH                 : num   8.36 8.26 8.94 8.24 8.26 8.32 8.4 8.38 8.38 8.5 ...
## $ Oxygen.Filler      : num   0.022 0.026 0.024 0.03 0.03 0.024 0.066 0.046 0.064 0.022 ...
## $ Bowl.Setpoint      : int  120 120 120 120 120 120 120 120 120 120 ...
## $ Pressure.Setpoint  : num   46.4 46.8 46.6 46 46 46 46 46 46 46 ...
## $ Air.Pressurer      : num   143 143 142 146 146 ...
## $ Alch.Rel           : num   6.58 6.56 7.66 7.14 7.14 7.16 6.54 6.52 6.52 6.54 ...
## $ Carb.Rel           : num   5.32 5.3 5.84 5.42 5.44 5.44 5.38 5.34 5.34 5.34 ...
## $ Balling.Lvl        : num   1.48 1.56 3.28 3.04 3.04 3.02 1.44 1.44 1.44 1.38 ...
```

All the rows with numbers are formatted as numbers. All the variables are numeric except for brand code, likely the only categorical variable.

1.1.1 Recap

- Loaded the training and evaluation sets from the project GitHub repo
- Inspected the structure with `glimpse()` and `str()`
- All variables came in as numeric except `Brand.Code`, which is the only categorical

1.2 Exploring the data

Recap located at the end of the section. Skip there to jump past the code/workflow.

We'll look at:

- The number of cases
- Missing values
- Data types (numerical vs. categorical)

1.2.1 Number of cases

```
nrow(training_data)
```

```
## [1] 2571
```

```
nrow(test_data)
```

```
## [1] 267
```

The training data has 2571 cases and the test data has 267.

1.2.2 Variables

Counting the number of columns/variables:

```
ncol(test_data)
```

```
## [1] 33
```

```
ncol(training_data)
```

```
## [1] 33
```

Both data sets have 33 columns.

Are all the variables represented in both data sets?

```
colnames(test_data)
```

```
## [1] "Brand.Code"      "Carb.Volume"      "Fill.Ounces"
## [4] "PC.Volume"       "Carb.Pressure"    "Carb.Temp"
## [7] "PSC"            "PSC.Fill"         "PSC.CO2"
## [10] "Mnf.Flow"        "Carb.Pressure1"   "Fill.Pressure"
## [13] "Hyd.Pressure1"   "Hyd.Pressure2"    "Hyd.Pressure3"
## [16] "Hyd.Pressure4"   "Filler.Level"     "Filler.Speed"
## [19] "Temperature"     "Usage.cont"       "Carb.Flow"
## [22] "Density"        "MFR"              "Balling"
## [25] "Pressure.Vacuum" "PH"               "Oxygen.Filler"
## [28] "Bowl.Setpoint"   "Pressure.Setpoint" "Air.Pressurer"
## [31] "Alch.Rel"        "Carb.Rel"         "Balling.Lvl"
```

```
colnames(training_data)
```

```
## [1] "Brand.Code"      "Carb.Volume"      "Fill.Ounces"
## [4] "PC.Volume"       "Carb.Pressure"    "Carb.Temp"
## [7] "PSC"            "PSC.Fill"         "PSC.CO2"
## [10] "Mnf.Flow"        "Carb.Pressure1"   "Fill.Pressure"
## [13] "Hyd.Pressure1"   "Hyd.Pressure2"    "Hyd.Pressure3"
## [16] "Hyd.Pressure4"   "Filler.Level"     "Filler.Speed"
## [19] "Temperature"     "Usage.cont"       "Carb.Flow"
## [22] "Density"        "MFR"              "Balling"
## [25] "Pressure.Vacuum" "PH"               "Oxygen.Filler"
## [28] "Bowl.Setpoint"   "Pressure.Setpoint" "Air.Pressurer"
## [31] "Alch.Rel"        "Carb.Rel"         "Balling.Lvl"
```

Both data sets have the same columns, so we can proceed.

1.2.3 Missing data

How many missing values are there? Are there any trends? How many complete cases are there?

1.2.3.1 Evaluation set (test_data)

Note: `test_data` here is the evaluation file used for the project deliverable. PH is intentionally blank in this file because that's what the model predicts. It's not a held-out test split from training.

Count nulls/missing values in the evaluation set:

```
sort(colSums(is.na(test_data)), decreasing = TRUE)
```

```
##           PH           MFR      Filler.Speed      Brand.Code
##          267           31           10           8
##      Fill.Ounces           PSC      PSC.CO2      PC.Volume
##           6           5           5           4
##      Carb.Pressure1      Hyd.Pressure4      PSC.Fill      Oxygen.Filler
##           4           4           3           3
##           Alch.Rel      Fill.Pressure      Filler.Level      Temperature
##           3           2           2           2
##           Usage.cont      Pressure.Setpoint      Carb.Rel      Carb.Volume
##           2           2           2           1
##           Carb.Temp      Hyd.Pressure2      Hyd.Pressure3      Density
##           1           1           1           1
##           Balling      Pressure.Vacuum      Bowl.Setpoint      Air.Pressurer
##           1           1           1           1
##           Carb.Pressure      Mnf.Flow      Hyd.Pressure1      Carb.Flow
##           0           0           0           0
##           Balling.Lvl
##           0
```

First off, all the **pH values are missing**. This is not a test set -it's what we will predict. The training data will have to be split on its own into a training and test set.

Most columns are missing a few (under 10) values or none at all. MFR is missing the most values - 31 out of 267, or about 12%. Filler speed is next, missing only 10 values.

Next, we'll look at the data row-wise and see if there are full or close-to-full missing cases.

```
rowSums(is.na(test_data))
```

```
## [1] 3 1 1 4 1 2 1 4 1 2 3 1 1 1 1 4 1 1 1 1 2 3 1 1
## [26] 1 1 1 2 1 1 2 2 2 2 1 1 1 1 2 1 2 1 1 1 1 1 1 1
## [51] 3 1 2 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [76] 1 2 1 1 1 1 1 1 1 1 5 2 1 1 1 1 2 15 1 2 2 1 3 1 2 1
## [101] 2 1 1 1 1 1 1 1 1 1 2 1 1 1 1 3 1 1 2 1 1 2 1 1 1 1
## [126] 1 2 2 1 1 2 2 2 3 1 1 3 2 2 2 1 1 1 2 1 1 1 1 2 1
## [151] 1 1 1 1 1 1 1 1 1 1 2 2 1 1 1 2 2 1 1 1 2 2 1 1 1
## [176] 1 1 1 2 1 1 1 1 1 2 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1
## [201] 1 1 2 1 1 1 1 2 2 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [226] 1 3 1 1 1 1 2 1 1 1 2 1 1 1 1 1 2 1 1 1 1 1 3 1
## [251] 2 1 3 1 1 9 2 1 1 1 1 1 1 1 1 1 1
```

```
#totals by row (1 does not count)
table(rowSums(is.na(test_data)))
```

```
##
##      1      2      3      4      5      9     15
## 200    50    11      3      1      1      1
```

```
#total NAs (subtracting 267 for the missing pH values)
sum(is.na(test_data)) - 267
```

```
## [1] 107
```

Overall, a data set with 267 rows and 33 columns is missing 107 values, or about 1.3%. 67 out of 267 cases are missing at least one value (about 25%).

One case has 14 missing values, so it may make sense to drop that; another is missing 8. Otherwise, most rows with missing values are missing only 1-3.

1.2.3.2 Training data Which variables are missing data?

```
sort(colSums(is.na(training_data)), decreasing = TRUE)
```

```
##           MFR           Brand.Code       Filler.Speed       PC.Volume
##           212             120           57             39
##      PSC.CO2      Fill.Ounces           PSC      Carb.Pressure1
##           39             38           33             32
##      Hyd.Pressure4      Carb.Pressure       Carb.Temp       PSC.Fill
##           30             27           26             23
##      Fill.Pressure      Filler.Level      Hyd.Pressure2      Hyd.Pressure3
##           22             20           15             15
##      Temperature      Oxygen.Filler      Pressure.Setpoint      Hyd.Pressure1
##           14             12           12             11
##      Carb.Volume      Carb.Rel           Alch.Rel           Usage.cont
##           10             10            9             5
##           PH           Mnf.Flow       Carb.Flow       Bowl.Setpoint
##            4             2            2             2
##      Density           Balling       Balling.Lvl      Pressure.Vacuum
##            1             1            1             0
##      Air.Pressurer
##            0
```

MFR is missing 212 out of 2571 cases, about 8.2%. Brand code, the only categorical variable, is missing 120 times, or about 4.7% of the time. Filler speed is missing 57 times, about 2.2%, and everything else is under 2%.

Let's see if anything strange is happening across rows:

```
#nothing out of the ordinary across rows. Most rows are complete.
#Rows with NAs typically 1 or 2 NAs, with
rowSums(is.na(training_data))
```

```

##      [1] 2 2 2 1 0 0 2 1 0 0 0 0 1 0 0 0 0 0 0 0 2 0 0
##      [25] 0 1 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 1 0 0 0 0 1
##      [49] 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0
##      [73] 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0
##      [97] 0 1 0 0 0 0 0 0 0 0 0 2 5 0 0 0 0 0 1 0 0 0 0
##     [121] 0 0 0 0 2 0 0 1 0 0 0 0 0 1 0 0 0 2 0 0 0 0 0
##     [145] 0 0 0 0 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##     [169] 0 0 0 0 0 0 0 0 1 0 2 0 0 0 0 0 2 0 0 0 0 0 0
##     [193] 0 0 0 1 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 0 0 0
##     [217] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 1 0 0 0 1
##     [241] 0 1 6 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 2
##     [265] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##     [289] 0 0 0 0 0 0 0 0 1 1 1 1 1 0 0 0 0 3 0 0 0 0 2
##     [313] 0 0 0 0 0 0 0 1 0 0 0 1 1 0 1 0 0 0 1 0 0 1 2
##     [337] 0 1 0 2 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0
##     [361] 0 0 0 0 2 2 0 2 0 0 0 1 0 2 1 0 1 0 0 0 0 0 0
##     [385] 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 2 0 0 0 0 0
##     [409] 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 2 2 1 0
##     [433] 0 0 2 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0
##     [457] 0 0 1 0 0 0 1 0 0 0 0 0 0 0 2 1 0 0 4 1 1 1 1
##     [481] 1 1 1 0 0 0 0 0 0 0 0 0 1 0 1 0 3 0 1 1 0 0 0
##     [505] 0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0
##     [529] 2 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1 0 0 0 1
##     [553] 0 0 0 0 0 1 0 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0
##     [577] 1 0 0 0 0 0 0 1 0 0 0 3 0 0 1 0 0 0 0 0 0 0 0
##     [601] 0 0 0 0 0 0 0 0 2 4 0 0 0 0 0 0 0 0 0 0 0 1 0
##     [625] 0 0 0 0 3 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 0 0
##     [649] 1 0 0 0 0 0 1 0 2 2 1 0 0 0 0 0 0 0 0 0 0 1 0
##     [673] 0 1 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 5 2 0 1 2
##     [697] 3 0 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 1 0 0 3
##     [721] 0 2 0 0 0 0 0 0 0 0 0 0 1 2 0 0 0 0 9 4 0 0 0
##     [745] 0 0 0 0 0 1 0 0 1 0 3 0 0 1 0 0 0 0 0 0 1 0 0
##     [769] 0 0 2 2 3 3 3 3 4 3 3 3 3 3 3 3 4 0 0 0 0 0 0
##     [793] 0 0 2 0 0 0 1 2 4 1 0 0 0 0 0 0 2 1 0 0 0 0 3
##     [817] 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 2 1 0 0 0 1 0
##     [841] 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##     [865] 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 2 0 0 0 0 1 0
##     [889] 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 6 0 0
##     [913] 2 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 0 0 0 0 1 0
##     [937] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 1 1 1 1
##     [961] 2 1 1 2 2 0 0 0 1 0 2 0 0 0 0 0 0 0 0 0 0 0 0
##     [985] 0 0 0 0 0 0 0 0 0 0 2 1 1 1 0 3 0 0 0 0 0 0 0
##    [1009] 0 0 0 0 0 2 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 2 0
##    [1033] 0 0 2 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0
##    [1057] 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0
##    [1081] 0 0 0 0 0 0 0 1 0 0 0 0 0 8 2 0 1 0 0 0 0 1 0
##    [1105] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1
##    [1129] 0 0 0 0 0 0 0 0 3 1 0 0 0 2 0 0 0 0 0 0 0 0 0
##    [1153] 0 0 0 0 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0
##    [1177] 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0
##    [1201] 1 0 1 0 0 0 0 0 1 1 1 1 3 2 3 2 4 1 1 1 1 0 0
##    [1225] 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 4 2 0 0 1 0
##    [1249] 0 0 0 0 3 0 0 0 0 0 0 2 0 0 0 0 1 0 0 1 1 1 0
##    [1273] 1 0 0 0 0 0 1 3 0 0 0 0 1 0 0 0 0 0 0 2 0 2 1

```



```

## [1297] 0 1 0 4 0 1 0 1 0 0 1 1 1 0 0 0 0 1 1 1 2 1 1
## [1321] 1 1 1 2 1 1 0 0 0 0 1 1 1 1 1 1 2 1 1 1 2 1 1
## [1345] 1 1 2 1 1 1 1 1 1 1 2 1 2 0 1 1 1 1 1 1 3 1 1
## [1369] 2 1 0 0 0 0 0 2 0 0 0 0 0 0 0 1 1 1 1 0 0 0 0
## [1393] 0 0 0 2 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [1417] 0 0 0 0 3 2 0 0 0 0 0 1 0 0 3 0 0 0 0 0 0 0 0
## [1441] 0 0 0 0 0 0 0 0 3 0 0 0 0 0 0 0 0 0 0 3 0 0 0
## [1465] 0 0 1 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0
## [1489] 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0
## [1513] 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 4 1 0 0 0 0 6 0
## [1537] 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 1 1 1 1
## [1561] 2 3 0 0 0 0 0 0 0 0 0 0 0 1 3 0 0 0 0 0 0 0 0
## [1585] 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [1609] 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0
## [1633] 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [1657] 0 0 0 0 0 0 0 0 1 1 0 0 0 0 1 0 1 0 0 0 0 1 0
## [1681] 0 0 1 0 0 1 1 0 1 1 0 0 0 0 0 0 0 0 0 0 0 1 0
## [1705] 0 0 1 0 0 0 0 0 0 0 0 0 0 0 12 1 0 0 1 0 3 1 2
## [1729] 0 0 0 1 0 0 0 2 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0
## [1753] 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 1 1 0 1 0
## [1777] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 2 1 1
## [1801] 4 1 1 1 1 1 3 1 0 0 0 0 0 0 0 2 2 0 0 0 0 0 0
## [1825] 2 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 5 0 0 0
## [1849] 0 0 1 0 0 2 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0
## [1873] 0 0 0 0 0 0 3 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0
## [1897] 0 0 0 2 0 2 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 1 0
## [1921] 1 0 0 0 0 2 0 1 0 0 1 0 0 0 0 0 0 0 1 0 1 0 0
## [1945] 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 2 1 1 1 1 0 0
## [1969] 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0
## [1993] 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1
## [2017] 3 1 1 1 3 1 1 3 1 1 0 0 0 0 0 1 0 0 0 0 0 0 0
## [2041] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [2065] 0 2 0 1 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 2 0 0
## [2089] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1 0
## [2113] 0 0 0 0 0 0 0 0 0 0 0 4 0 0 0 0 1 0 0 0 0 0 0
## [2137] 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 2 0 0
## [2161] 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0
## [2185] 1 1 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0
## [2209] 0 0 0 0 0 5 0 0 0 1 0 0 0 0 0 0 2 0 0 0 0 0 0
## [2233] 0 0 1 0 0 0 0 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0
## [2257] 2 1 1 1 0 0 0 0 0 0 0 0 0 0 4 0 0 0 0 0 0 2 0
## [2281] 0 1 0 0 0 0 1 0 0 0 0 2 0 0 0 0 1 0 0 0 2 0 0
## [2305] 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 1 2
## [2329] 0 1 1 0 1 0 0 0 0 0 1 2 0 0 0 0 0 0 0 0 0 0 0
## [2353] 0 0 0 0 2 0 3 0 0 0 0 0 0 0 0 0 1 0 0 3 0 0 0
## [2377] 0 0 0 2 0 0 0 0 2 3 0 0 1 0 0 0 0 0 1 0 0 0 0
## [2401] 0 0 0 0 0 0 0 0 1 0 0 0 2 2 0 0 0 0 0 0 0 0 0
## [2425] 0 0 1 1 1 0 0 0 0 0 0 2 0 1 0 0 0 0 0 0 0 1 0
## [2449] 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 1 1
## [2473] 0 0 0 0 2 0 2 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0
## [2497] 0 0 0 0 0 0 0 1 0 0 0 7 1 0 0 0 0 2 0 0 0 0 0
## [2521] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
## [2545] 0 0 0 0 0 1 0 0 0 0 0 0 0 0 2 0 0 0 1 0 0 1 1
## [2569] 0 0 3

```

```
table(rowSums(is.na(training_data)))
```

```
##
##      0      1      2      3      4      5      6      7      8      9     12
## 2038  345  119   45   13    4    3    1    1    1    1
```

```
sum(is.na(test_data))
```

```
## [1] 374
```

Overall, 844 values are missing across the training set, about 1% of all cells. 2038/2571 cases are complete. That's about 79% complete, 21% incomplete. Let's also check to see if the data is MCAR.

```
library(naniar)
numeric_only <- training_data %>%
  select(-Brand.Code)

numeric_only <- numeric_only %>%
  dplyr::select(where(is.numeric)) %>%
  mutate(across(everything(), as.double))

mcar_test(numeric_only)
```

```
## # A tibble: 1 x 4
##   statistic    df p.value missing.patterns
##   <dbl> <dbl> <dbl>         <int>
## 1  10791.  2871     0             99
```

The p-value is 0, which means the data most likely isn't missing completely at random (there are patterns in the missing data).

1.2.3.3 A closer look at Brand.Code missingness The training set has 120 rows where Brand.Code is blank (about 4.7% of cases). Before deciding how to handle them in modeling, we check whether those rows look different from the rest. If the missingness carries signal about PH, we should preserve it rather than impute it away.

```
# How does PH look in rows where Brand.Code is missing vs known?
training_data %>%
  mutate(brand_missing = is.na(Brand.Code) | Brand.Code == "") %>%
  filter(!is.na(PH)) %>%
  group_by(brand_missing) %>%
  summarise(
    n = n(),
    ph_mean = mean(PH),
    ph_median = median(PH),
    ph_sd = sd(PH),
    ph_min = min(PH),
    ph_max = max(PH)
  )
```

```
## # A tibble: 2 x 7
##   brand_missing      n ph_mean ph_median ph_sd ph_min ph_max
##   <lgl>          <int>   <dbl>   <dbl> <dbl> <dbl> <dbl>
## 1 FALSE         2447    8.55    8.54 0.172   7.88   9.36
## 2 TRUE          120    8.49    8.51 0.177   8.12   8.88
```

Two visible differences. Missing-Brand rows have a mean PH about 0.06 units below known-Brand rows (8.49 vs 8.55), and a noticeably narrower PH range (8.12 to 8.88 vs 7.88 to 9.36). They cluster toward the middle of the distribution rather than the tails.

That looks like what Kuhn and Johnson calls informative missingness: the absence of a Brand.Code value is itself associated with a specific PH pattern. Imputing the missing values into one of A/B/C/D would roll these rows into whichever brand the imputer picks, distorting that brand's signal and erasing the mean shift. The cleaner approach is to keep the missingness as its own level ("Unknown") when we model.

1.2.4 Recap

- 2571 training cases, 267 evaluation cases, 33 columns in both
- All PH values in the evaluation set are missing. That file is what we predict on, not a test set
- Missingness concentrates in **MFR** (~8% training, ~12% eval), **Brand.Code** (~5% training), and **Filler.Speed** (~2% training); everything else under 2%
- ~79% of training cases are complete; the MCAR test (`naniar::mcar_test`) returned $p \sim 0$, so missingness isn't completely at random
- A closer look at Brand.Code's missingness shows the 120 missing-Brand rows have a slightly lower mean PH and a narrower PH range than known-Brand rows, so the missingness itself looks informative; we'll preserve it rather than impute when modeling

1.3 Data cleaning

Recap located at the end of the section. Skip there to jump past the code/workflow.

We'll cover:

- Variables with zero or near-zero variance
- Highly correlated variables
- Outliers/range of variables/scaling (if needed)

1.3.1 Checking for variables with near-zero or zero variance

This data set has 33 variables. Let's check to see if they're all (potentially) providing useful predictive information by testing for near-zero or zero variance.

```
library(caret)
nzv_results <- nearZeroVar(training_data, saveMetrics = TRUE)

nzv_results
```

```
##           freqRatio percentUnique zeroVar   nzv
## Brand.Code    2.014634     0.1555815  FALSE FALSE
## Carb.Volume   1.055556     3.9284325  FALSE FALSE
```

## Fill.Ounces	1.163043	3.5783742	FALSE	FALSE
## PC.Volume	1.100000	17.6584986	FALSE	FALSE
## Carb.Pressure	1.016393	4.1229094	FALSE	FALSE
## Carb.Temp	1.016667	4.7841307	FALSE	FALSE
## PSC	1.211538	5.0175029	FALSE	FALSE
## PSC.Fill	1.091892	1.2446519	FALSE	FALSE
## PSC.CO2	1.078303	0.5056398	FALSE	FALSE
## Mnf.Flow	1.048443	18.9420459	FALSE	FALSE
## Carb.Pressure1	1.031746	5.4453520	FALSE	FALSE
## Fill.Pressure	1.762931	4.2007001	FALSE	FALSE
## Hyd.Pressure1	31.111111	9.5293660	FALSE	TRUE
## Hyd.Pressure2	7.271028	8.0513419	FALSE	FALSE
## Hyd.Pressure3	11.450704	7.4679113	FALSE	FALSE
## Hyd.Pressure4	1.008264	1.5558149	FALSE	FALSE
## Filler.Level	1.086957	11.2018670	FALSE	FALSE
## Filler.Speed	1.045161	9.4904706	FALSE	FALSE
## Temperature	1.040000	2.1781408	FALSE	FALSE
## Usage.cont	1.105263	18.7086737	FALSE	FALSE
## Carb.Flow	1.586207	20.7312330	FALSE	FALSE
## Density	1.108374	3.0338390	FALSE	FALSE
## MFR	1.222222	22.8315830	FALSE	FALSE
## Balling	1.193182	8.4402956	FALSE	FALSE
## Pressure.Vacuum	1.389728	0.6223259	FALSE	FALSE
## PH	1.078125	2.0225593	FALSE	FALSE
## Oxygen.Filler	1.266667	13.1466356	FALSE	FALSE
## Bowl.Setpoint	2.990847	0.4278491	FALSE	FALSE
## Pressure.Setpoint	1.319361	0.3111630	FALSE	FALSE
## Air.Pressurer	1.093407	1.2446519	FALSE	FALSE
## Alch.Rel	1.098315	2.0614547	FALSE	FALSE
## Carb.Rel	1.011583	1.6336056	FALSE	FALSE
## Balling.Lvl	1.294872	3.1894205	FALSE	FALSE

Hyd.Pressure1 has a near-zero variance, so we could eliminate that one.

1.3.2 Looking for highly correlated variables

We'll create a correlation grid, so we can see actual correlation between variables. We'll also create a correlation plot and use `findCorrelation` from the `caret` package to find variables that have 90% or higher correlation.

```
correlation_grid <- training_data %>%
  drop_na() %>%
  select(-Brand.Code) %>%
  cor()

correlation_grid <- as.data.frame(correlation_grid)

correlation_grid
```

##	Carb.Volume	Fill.Ounces	PC.Volume	Carb.Pressure
## Carb.Volume	1.000000000	0.022648036	-0.243234517	0.4278586099
## Fill.Ounces	0.022648036	1.000000000	-0.172701966	0.0044669914
## PC.Volume	-0.243234517	-0.172701966	1.000000000	-0.1300941804

## Carb.Pressure	0.427858610	0.004466991	-0.130094180	1.0000000000
## Carb.Temp	-0.135080260	-0.020073097	0.008599427	0.8243350333
## PSC	-0.020625187	0.043056041	0.190961266	-0.0520086907
## PSC.Fill	-0.012364141	0.069748653	-0.055914847	-0.0179275381
## PSC.CO2	-0.051977547	0.014705610	-0.055356933	-0.0174011136
## Mnf.Flow	0.100525364	-0.005787251	-0.203032882	0.0189865815
## Carb.Pressure1	0.081146212	-0.004186326	-0.238496181	0.0221844474
## Fill.Pressure	-0.096123314	0.068377098	-0.095744421	-0.0850458606
## Hyd.Pressure1	-0.051894268	-0.146359721	0.271171741	-0.0704043564
## Hyd.Pressure2	0.033567314	-0.143047986	0.028768135	-0.0267152675
## Hyd.Pressure3	0.057610071	-0.107816452	-0.024741466	-0.0082140513
## Hyd.Pressure4	-0.567908627	0.056086557	0.132413786	-0.3229128556
## Filler.Level	-0.018377226	0.010856355	0.220434988	0.0097134297
## Filler.Speed	0.003233452	0.044277046	-0.073766021	0.0320230919
## Temperature	-0.189760851	-0.011208233	0.142065435	-0.0490025241
## Usage.cont	0.073297402	0.085289046	-0.269204705	0.0004415267
## Carb.Flow	-0.088782919	-0.116991410	0.254535534	-0.0111938869
## Density	0.801808803	-0.080414770	-0.167310345	0.4591743002
## MFR	0.010144634	0.043445999	-0.066678296	0.0292986178
## Balling	0.818778791	-0.068439337	-0.204785622	0.4539307252
## Pressure.Vacuum	-0.079190180	0.052165821	-0.051605143	0.0051279147
## PH	0.073251595	-0.118207504	0.097746859	0.0788617427
## Oxygen.Filler	-0.112682435	-0.032064258	0.212285069	-0.0506609705
## Bowl.Setpoint	0.001993607	0.011188522	0.210972310	0.0187581570
## Pressure.Setpoint	-0.160640146	0.065078231	-0.022210124	-0.1171228371
## Air.Pressurer	-0.024854172	0.058101103	-0.036605483	0.0219237613
## Alch.Rel	0.824787575	-0.110432476	-0.177865067	0.4590380468
## Carb.Rel	0.841072250	-0.118666586	-0.167433257	0.4736224720
## Balling.Lvl	0.817993347	-0.063820071	-0.210081156	0.4589466196
##	Carb.Temp	PSC	PSC.Fill	PSC.CO2
## Carb.Volume	-0.135080260	-0.020625187	-0.012364141	-0.051977547
## Fill.Ounces	-0.020073097	0.043056041	0.069748653	0.014705610
## PC.Volume	0.008599427	0.190961266	-0.055914847	-0.055356933
## Carb.Pressure	0.824335033	-0.052008691	-0.017927538	-0.017401114
## Carb.Temp	1.000000000	-0.044380293	-0.010968792	0.023660514
## PSC	-0.044380293	1.000000000	0.193242059	0.053922116
## PSC.Fill	-0.010968792	0.193242059	1.000000000	0.201967429
## PSC.CO2	0.023660514	0.053922116	0.201967429	1.000000000
## Mnf.Flow	-0.030110321	0.042982180	-0.033801564	0.049931986
## Carb.Pressure1	-0.014312694	-0.011071066	-0.029772433	0.038316144
## Fill.Pressure	-0.028433189	0.030908100	-0.010528126	0.078519840
## Hyd.Pressure1	-0.045431217	-0.019758828	-0.053220514	-0.049037869
## Hyd.Pressure2	-0.039543090	-0.033813839	-0.080467321	-0.021838338
## Hyd.Pressure3	-0.035120030	-0.020448188	-0.068958062	0.005757630
## Hyd.Pressure4	-0.002573081	0.036485173	0.019810466	0.054940092
## Filler.Level	0.011038984	-0.016061814	0.048227359	-0.042306839
## Filler.Speed	0.029355933	-0.002395360	-0.017295907	-0.008421406
## Temperature	0.061914927	0.014217909	0.030080929	0.037007308
## Usage.cont	-0.043584359	0.049860720	-0.025727952	0.032502663
## Carb.Flow	0.045050619	-0.037361416	0.017579095	-0.013058433
## Density	0.021593836	-0.067168725	-0.024708871	-0.045382139
## MFR	0.023442657	0.007659111	-0.008663604	-0.004263878
## Balling	0.006005105	-0.064450826	-0.019108545	-0.043036807
## Pressure.Vacuum	0.037327836	0.055246194	0.033093793	-0.018438819

## PH	0.034594375	-0.069906559	-0.031374041	-0.088759922
## Oxygen.Filler	0.012010201	0.002765929	-0.005769535	-0.016971816
## Bowl.Setpoint	0.007550616	-0.020808347	0.049191096	-0.044820225
## Pressure.Setpoint	-0.027029545	0.042979144	-0.005800041	0.082772777
## Air.Pressurer	0.038177076	0.061433424	-0.017995701	-0.006984004
## Alch.Rel	0.004966509	-0.060809035	-0.017045886	-0.050859393
## Carb.Rel	0.009020802	-0.065354969	-0.015365172	-0.066475871
## Balling.Lvl	0.010458674	-0.061576860	-0.016064731	-0.047296997
##	Mnf.Flow	Carb.Pressure1	Fill.Pressure	Hyd.Pressure1
## Carb.Volume	0.100525364	0.081146212	-0.09612331	-0.051894268
## Fill.Ounces	-0.005787251	-0.004186326	0.06837710	-0.146359721
## PC.Volume	-0.203032882	-0.238496181	-0.09574442	0.271171741
## Carb.Pressure	0.018986582	0.022184447	-0.08504586	-0.070404356
## Carb.Temp	-0.030110321	-0.014312694	-0.02843319	-0.045431217
## PSC	0.042982180	-0.011071066	0.03090810	-0.019758828
## PSC.Fill	-0.033801564	-0.029772433	-0.01052813	-0.053220514
## PSC.CO2	0.049931986	0.038316144	0.07851984	-0.049037869
## Mnf.Flow	1.000000000	0.465355315	0.49111329	0.298997690
## Carb.Pressure1	0.465355315	1.000000000	0.22002705	-0.012711716
## Fill.Pressure	0.491113286	0.220027054	1.00000000	0.137594249
## Hyd.Pressure1	0.298997690	-0.012711716	0.13759425	1.000000000
## Hyd.Pressure2	0.648337926	0.284710255	0.31836940	0.711851345
## Hyd.Pressure3	0.764142833	0.375658047	0.43637028	0.605589019
## Hyd.Pressure4	0.073124200	0.038936494	0.29144885	0.051658157
## Filler.Level	-0.608380668	-0.425646963	-0.45892889	-0.029684260
## Filler.Speed	0.024363769	0.012389362	-0.21347499	-0.022850088
## Temperature	-0.063477350	-0.070067643	0.06397894	-0.073341745
## Usage.cont	0.560192617	0.354875344	0.30257079	0.118691986
## Carb.Flow	-0.497211539	-0.273632871	-0.18715299	-0.169136801
## Density	0.021343082	0.040888366	-0.20887500	-0.003601856
## MFR	0.019593355	0.013931715	-0.21290076	-0.035646540
## Balling	0.112393982	0.073508154	-0.15676227	0.025863408
## Pressure.Vacuum	-0.559943315	-0.284881843	-0.30810832	-0.324133005
## PH	-0.457218313	-0.112108310	-0.31311751	-0.048631533
## Oxygen.Filler	-0.550066399	-0.278795209	-0.24769592	-0.118317983
## Bowl.Setpoint	-0.611580420	-0.432671459	-0.42609290	-0.029844050
## Pressure.Setpoint	0.482530739	0.234201877	0.81528556	0.190079720
## Air.Pressurer	-0.041525175	0.061506659	0.03280945	-0.209227461
## Alch.Rel	0.027999719	0.009047055	-0.20467008	0.006667887
## Carb.Rel	-0.027880871	0.004804776	-0.20272201	0.034387945
## Balling.Lvl	0.042267589	0.033601804	-0.17885789	-0.007754220
##	Hyd.Pressure2	Hyd.Pressure3	Hyd.Pressure4	Filler.Level
## Carb.Volume	0.03356731	0.057610071	-0.567908627	-0.01837723
## Fill.Ounces	-0.14304799	-0.107816452	0.056086557	0.01085635
## PC.Volume	0.02876814	-0.024741466	0.132413786	0.22043499
## Carb.Pressure	-0.02671527	-0.008214051	-0.322912856	0.00971343
## Carb.Temp	-0.03954309	-0.035120030	-0.002573081	0.01103898
## PSC	-0.03381384	-0.020448188	0.036485173	-0.01606181
## PSC.Fill	-0.08046732	-0.068958062	0.019810466	0.04822736
## PSC.CO2	-0.02183834	0.005757630	0.054940092	-0.04230684
## Mnf.Flow	0.64833793	0.764142833	0.073124200	-0.60838067
## Carb.Pressure1	0.28471026	0.375658047	0.038936494	-0.42564696
## Fill.Pressure	0.31836940	0.436370279	0.291448848	-0.45892889
## Hyd.Pressure1	0.71185135	0.605589019	0.051658157	-0.02968426

## Hyd.Pressure2	1.00000000	0.917661875	0.022980025	-0.41103521
## Hyd.Pressure3	0.91766187	1.000000000	0.046649319	-0.52796590
## Hyd.Pressure4	0.02298003	0.046649319	1.000000000	-0.10010344
## Filler.Level	-0.41103521	-0.527965898	-0.100103442	1.00000000
## Filler.Speed	0.07028157	0.031692930	-0.259685780	0.06319357
## Temperature	-0.13725053	-0.120288087	0.283933984	0.05040323
## Usage.cont	0.37224053	0.419761856	0.070916924	-0.35740823
## Carb.Flow	-0.32187768	-0.363691033	-0.009527157	0.01743320
## Density	0.05842429	0.040354370	-0.584381995	0.00772950
## MFR	0.04937645	0.012841039	-0.280014344	0.06935680
## Balling	0.10480326	0.125869294	-0.594996647	0.01188558
## Pressure.Vacuum	-0.62816754	-0.672258326	-0.054652623	0.35866815
## PH	-0.21571713	-0.260535183	-0.184518087	0.34879540
## Oxygen.Filler	-0.29438355	-0.363660021	0.009692254	0.22540536
## Bowl.Setpoint	-0.41457787	-0.529215855	-0.117019292	0.97724166
## Pressure.Setpoint	0.34183180	0.459918732	0.288770453	-0.44595292
## Air.Pressurer	-0.17237591	-0.067700529	0.021731381	-0.12573419
## Alch.Rel	0.03916606	0.049948952	-0.685576013	0.04355856
## Carb.Rel	0.03150549	0.013543248	-0.581827500	0.13107856
## Balling.Lvl	0.02885940	0.041595480	-0.575044299	0.05317923
##	Filler.Speed	Temperature	Usage.cont	Carb.Flow
## Carb.Volume	0.003233452	-0.18976085	0.0732974017	-0.088782919
## Fill.Ounces	0.044277046	-0.01120823	0.0852890456	-0.116991410
## PC.Volume	-0.073766021	0.14206544	-0.2692047046	0.254535534
## Carb.Pressure	0.032023092	-0.04900252	0.0004415267	-0.011193887
## Carb.Temp	0.029355933	0.06191493	-0.0435843590	0.045050619
## PSC	-0.002395360	0.01421791	0.0498607203	-0.037361416
## PSC.Fill	-0.017295907	0.03008093	-0.0257279520	0.017579095
## PSC.CO2	-0.008421406	0.03700731	0.0325026634	-0.013058433
## Mnf.Flow	0.024363769	-0.06347735	0.5601926171	-0.497211539
## Carb.Pressure1	0.012389362	-0.07006764	0.3548753438	-0.273632871
## Fill.Pressure	-0.213474995	0.06397894	0.3025707885	-0.187152994
## Hyd.Pressure1	-0.022850088	-0.07334175	0.1186919862	-0.169136801
## Hyd.Pressure2	0.070281568	-0.13725053	0.3722405332	-0.321877677
## Hyd.Pressure3	0.031692930	-0.12028809	0.4197618556	-0.363691033
## Hyd.Pressure4	-0.259685780	0.28393398	0.0709169237	-0.009527157
## Filler.Level	0.063193573	0.05040323	-0.3574082260	0.017433196
## Filler.Speed	1.000000000	-0.06794451	0.0466749382	-0.062641899
## Temperature	-0.067944506	1.00000000	-0.0950428265	0.140119431
## Usage.cont	0.046674938	-0.09504283	1.0000000000	-0.488854072
## Carb.Flow	-0.062641899	0.14011943	-0.4888540721	1.000000000
## Density	0.025454377	-0.19369890	-0.0045531798	0.023058872
## MFR	0.951264445	-0.08156825	0.0397443802	-0.074148067
## Balling	0.037938064	-0.23844781	0.0692694347	-0.102299824
## Pressure.Vacuum	0.045187804	0.04313148	-0.3196915871	0.290026894
## PH	-0.047596639	-0.18152910	-0.3577042377	0.240757744
## Oxygen.Filler	-0.044126285	0.09338308	-0.3164959408	0.385692082
## Bowl.Setpoint	0.027688288	0.05010937	-0.3584563273	0.012221618
## Pressure.Setpoint	-0.068525700	0.05904194	0.2626733751	-0.166362628
## Air.Pressurer	-0.006199674	0.06692883	-0.1025056258	0.137849013
## Alch.Rel	-0.005598672	-0.24776232	-0.0237408078	0.007527013
## Carb.Rel	0.005300760	-0.13595194	-0.0390798757	-0.006940815
## Balling.Lvl	0.003823258	-0.22497504	0.0232069975	-0.055365859
##	Density	MFR	Balling	Pressure.Vacuum

## Carb.Volume	0.801808803	0.010144634	0.818778791	-0.079190180
## Fill.Ounces	-0.080414770	0.043445999	-0.068439337	0.052165821
## PC.Volume	-0.167310345	-0.066678296	-0.204785622	-0.051605143
## Carb.Pressure	0.459174300	0.029298618	0.453930725	0.005127915
## Carb.Temp	0.021593836	0.023442657	0.006005105	0.037327836
## PSC	-0.067168725	0.007659111	-0.064450826	0.055246194
## PSC.Fill	-0.024708871	-0.008663604	-0.019108545	0.033093793
## PSC.CO2	-0.045382139	-0.004263878	-0.043036807	-0.018438819
## Mnf.Flow	0.021343082	0.019593355	0.112393982	-0.559943315
## Carb.Pressure1	0.040888366	0.013931715	0.073508154	-0.284881843
## Fill.Pressure	-0.208875002	-0.212900763	-0.156762274	-0.308108315
## Hyd.Pressure1	-0.003601856	-0.035646540	0.025863408	-0.324133005
## Hyd.Pressure2	0.058424287	0.049376448	0.104803262	-0.628167539
## Hyd.Pressure3	0.040354370	0.012841039	0.125869294	-0.672258326
## Hyd.Pressure4	-0.584381995	-0.280014344	-0.594996647	-0.054652623
## Filler.Level	0.007729500	0.069356797	0.011885578	0.358668150
## Filler.Speed	0.025454377	0.951264445	0.037938064	0.045187804
## Temperature	-0.193698895	-0.081568252	-0.238447810	0.043131483
## Usage.cont	-0.004553180	0.039744380	0.069269435	-0.319691587
## Carb.Flow	0.023058872	-0.074148067	-0.102299824	0.290026894
## Density	1.000000000	0.039383665	0.953127632	-0.083762802
## MFR	0.039383665	1.000000000	0.051171065	0.043987724
## Balling	0.953127632	0.051171065	1.000000000	-0.167052546
## Pressure.Vacuum	-0.083762802	0.043987724	-0.167052546	1.000000000
## PH	0.095340535	-0.052377217	0.072297357	0.213173584
## Oxygen.Filler	-0.046621720	-0.048876320	-0.113757056	0.278478283
## Bowl.Setpoint	0.019792536	0.042312091	0.027415802	0.358696671
## Pressure.Setpoint	-0.250884909	-0.064390559	-0.210098742	-0.298520301
## Air.Pressurer	-0.082976446	-0.008182360	-0.102307854	0.175090738
## Alch.Rel	0.922993173	-0.001003949	0.945725964	-0.054788103
## Carb.Rel	0.860322233	0.010884909	0.858990647	-0.008434181
## Balling.Lvl	0.956417297	0.016039036	0.988412082	-0.051626926
##	PH	Oxygen.Filler	Bowl.Setpoint	Pressure.Setpoint
## Carb.Volume	0.073251595	-0.112682435	0.001993607	-0.160640146
## Fill.Ounces	-0.118207504	-0.032064258	0.011188522	0.065078231
## PC.Volume	0.097746859	0.212285069	0.210972310	-0.022210124
## Carb.Pressure	0.078861743	-0.050660970	0.018758157	-0.117122837
## Carb.Temp	0.034594375	0.012010201	0.007550616	-0.027029545
## PSC	-0.069906559	0.002765929	-0.020808347	0.042979144
## PSC.Fill	-0.031374041	-0.005769535	0.049191096	-0.005800041
## PSC.CO2	-0.088759922	-0.016971816	-0.044820225	0.082772777
## Mnf.Flow	-0.457218313	-0.550066399	-0.611580420	0.482530739
## Carb.Pressure1	-0.112108310	-0.278795209	-0.432671459	0.234201877
## Fill.Pressure	-0.313117513	-0.247695921	-0.426092896	0.815285564
## Hyd.Pressure1	-0.048631533	-0.118317983	-0.029844050	0.190079720
## Hyd.Pressure2	-0.215717133	-0.294383550	-0.414577866	0.341831797
## Hyd.Pressure3	-0.260535183	-0.363660021	-0.529215855	0.459918732
## Hyd.Pressure4	-0.184518087	0.009692254	-0.117019292	0.288770453
## Filler.Level	0.348795399	0.225405359	0.977241656	-0.445952917
## Filler.Speed	-0.047596639	-0.044126285	0.027688288	-0.068525700
## Temperature	-0.181529097	0.093383081	0.050109366	0.059041937
## Usage.cont	-0.357704238	-0.316495941	-0.358456327	0.262673375
## Carb.Flow	0.240757744	0.385692082	0.012221618	-0.166362628
## Density	0.095340535	-0.046621720	0.019792536	-0.250884909

## MFR	-0.052377217	-0.048876320	0.042312091	-0.064390559
## Balling	0.072297357	-0.113757056	0.027415802	-0.210098742
## Pressure.Vacuum	0.213173584	0.278478283	0.358696671	-0.298520301
## PH	1.000000000	0.162016479	0.358504933	-0.316950863
## Oxygen.Filler	0.162016479	1.000000000	0.224734144	-0.241221158
## Bowl.Setpoint	0.358504933	0.224734144	1.000000000	-0.440555036
## Pressure.Setpoint	-0.316950863	-0.241221158	-0.440555036	1.000000000
## Air.Pressurer	-0.008276544	0.116495267	-0.130939728	0.073805550
## Alch.Rel	0.156053375	-0.050111016	0.063332637	-0.268800293
## Carb.Rel	0.189575611	-0.039206163	0.152117160	-0.261257572
## Balling.Lvl	0.103279163	-0.074544984	0.070442468	-0.242343282
##	Air.Pressurer	Alch.Rel	Carb.Rel	Balling.Lvl
## Carb.Volume	-0.024854172	0.824787575	0.841072250	0.817993347
## Fill.Ounces	0.058101103	-0.110432476	-0.118666586	-0.063820071
## PC.Volume	-0.036605483	-0.177865067	-0.167433257	-0.210081156
## Carb.Pressure	0.021923761	0.459038047	0.473622472	0.458946620
## Carb.Temp	0.038177076	0.004966509	0.009020802	0.010458674
## PSC	0.061433424	-0.060809035	-0.065354969	-0.061576860
## PSC.Fill	-0.017995701	-0.017045886	-0.015365172	-0.016064731
## PSC.CO2	-0.006984004	-0.050859393	-0.066475871	-0.047296997
## Mnf.Flow	-0.041525175	0.027999719	-0.027880871	0.042267589
## Carb.Pressure1	0.061506659	0.009047055	0.004804776	0.033601804
## Fill.Pressure	0.032809452	-0.204670077	-0.202722006	-0.178857893
## Hyd.Pressure1	-0.209227461	0.006667887	0.034387945	-0.007754220
## Hyd.Pressure2	-0.172375910	0.039166055	0.031505491	0.028859401
## Hyd.Pressure3	-0.067700529	0.049948952	0.013543248	0.041595480
## Hyd.Pressure4	0.021731381	-0.685576013	-0.581827500	-0.575044299
## Filler.Level	-0.125734192	0.043558564	0.131078561	0.053179225
## Filler.Speed	-0.006199674	-0.005598672	0.005300760	0.003823258
## Temperature	0.066928829	-0.247762315	-0.135951942	-0.224975043
## Usage.cont	-0.102505626	-0.023740808	-0.039079876	0.023206997
## Carb.Flow	0.137849013	0.007527013	-0.006940815	-0.055365859
## Density	-0.082976446	0.922993173	0.860322233	0.956417297
## MFR	-0.008182360	-0.001003949	0.010884909	0.016039036
## Balling	-0.102307854	0.945725964	0.858990647	0.988412082
## Pressure.Vacuum	0.175090738	-0.054788103	-0.008434181	-0.051626926
## PH	-0.008276544	0.156053375	0.189575611	0.103279163
## Oxygen.Filler	0.116495267	-0.050111016	-0.039206163	-0.074544984
## Bowl.Setpoint	-0.130939728	0.063332637	0.152117160	0.070442468
## Pressure.Setpoint	0.073805550	-0.268800293	-0.261257572	-0.242343282
## Air.Pressurer	1.000000000	-0.082451451	-0.107386102	-0.085434755
## Alch.Rel	-0.082451451	1.000000000	0.882791450	0.947343341
## Carb.Rel	-0.107386102	0.882791450	1.000000000	0.872148303
## Balling.Lvl	-0.085434755	0.947343341	0.872148303	1.000000000

There are many highly correlated variables. A quick scan shows:

- hyd.pressure 2 and 3 have a .91 correlation
- filler.level and bowl.set.point have a .98 correlation
- fillers.speed and MFR have a .95 correlation
- density and balling.lvl have a .95 correlation


```
#if we remove colinear variables at 95% correlation, we would drop 10 variables
indexestodrop95 <- findCorrelation(cor(grid), cutoff = 0.95)
colnames(grid)[indexestodrop95]
```

```
## [1] "Pressure.Setpoint" "Mnf.Flow"          "Hyd.Pressure3"
## [4] "Bowl.Setpoint"     "Carb.Rel"          "Balling"
## [7] "Alch.Rel"          "Balling.Lvl"       "Density"
## [10] "Filler.Speed"
```

```
#this will leave us with 22 predictor variables and the response variable.
```

```
#if we remove colinear variables at 90% correlation, we would drop 12 variables
indexestodrop90 <- findCorrelation(cor(grid), cutoff = 0.90)
colnames(grid)[indexestodrop90]
```

```
## [1] "Pressure.Setpoint" "Mnf.Flow"          "Hyd.Pressure3"
## [4] "Pressure.Vacuum"   "Hyd.Pressure4"     "Bowl.Setpoint"
## [7] "Carb.Rel"          "Balling"           "Alch.Rel"
## [10] "Balling.Lvl"       "Density"           "Filler.Speed"
```

```
#this will leave us with 20 predictor variables and the response variable.
```

1.3.3 Removing variables

Removing some of the 33 variables will help simplify this data set. We will remove the variable with NZV and highly colinear variables (greater than or about .90, as defined above).

the findcorrelation function says to drop Filler.Speed instead of MFR. However, we will opt to drop MFR instead, given that it has the highest number of missing values (filler.speed has 57 as opposed to MFR's 212). Having to impute all these values will makes the variable less reliable, and dropping it would mean having to drop more cases.

```
#removing the variable with NZV
training_data_new <- training_data %>%
  select(-Hyd.Pressure1)

#removing correlated variables
training_data_new <- training_data_new %>%
  select(-Pressure.Setpoint, -MFR, -Hyd.Pressure3,
        -Pressure.Vacuum, -Hyd.Pressure4, -Bowl.Setpoint, -Carb.Rel,
        -Balling, -Alch.Rel, -Balling.Lvl, -Density)

#donig the same for the test set
test_data_new <- test_data %>%
  select(-Hyd.Pressure1, -Pressure.Setpoint, -MFR, -Hyd.Pressure3,
        -Pressure.Vacuum, -Hyd.Pressure4, -Bowl.Setpoint, -Carb.Rel,
        -Balling, -Alch.Rel, -Balling.Lvl, -Density)
```

1.3.4 Checking for outliers

Before we deal with missing values, let's see if there are any glaring outliers or data entry errors.

#may be an outlier in oxygen filler - the max calues is much higher than the

`summary(test_data_new)`

```
## Brand.Code      Carb.Volume      Fill.Ounces      PC.Volume
## Length:267      Min.      :5.147      Min.      :23.75      Min.      :0.09867
## Class :character 1st Qu.:5.287      1st Qu.:23.92      1st Qu.:0.23333
## Mode  :character Median :5.340      Median :23.97      Median :0.27533
##              Mean  :5.369      Mean  :23.97      Mean  :0.27769
##              3rd Qu.:5.465      3rd Qu.:24.01      3rd Qu.:0.32200
##              Max.   :5.667      Max.   :24.20      Max.   :0.46400
##              NA's   :1          NA's   :6          NA's   :4
## Carb.Pressure    Carb.Temp      PSC              PSC.Fill
## Min.      :60.20      Min.      :130.0      Min.      :0.00400      Min.      :0.0200
## 1st Qu.:65.30      1st Qu.:138.4      1st Qu.:0.04450      1st Qu.:0.1000
## Median :68.00      Median :140.8      Median :0.07600      Median :0.1800
## Mean      :68.25      Mean      :141.2      Mean      :0.08545      Mean      :0.1903
## 3rd Qu.:70.60      3rd Qu.:143.8      3rd Qu.:0.11200      3rd Qu.:0.2600
## Max.      :77.60      Max.      :154.0      Max.      :0.24600      Max.      :0.6200
##              NA's      :1          NA's      :5          NA's      :3
## PSC.CO2          Mnf.Flow      Carb.Pressure1    Fill.Pressure
## Min.      :0.00000      Min.      : -100.20      Min.      :113.0      Min.      :37.80
## 1st Qu.:0.02000      1st Qu.: -100.00      1st Qu.:120.2      1st Qu.:46.00
## Median :0.04000      Median :    0.20      Median :123.4      Median :47.80
## Mean      :0.05107      Mean      : 21.03      Mean      :123.0      Mean      :48.14
## 3rd Qu.:0.06000      3rd Qu.: 141.30      3rd Qu.:125.5      3rd Qu.:50.20
## Max.      :0.24000      Max.      : 220.40      Max.      :136.0      Max.      :60.20
## NA's      :5          NA's      :4          NA's      :2
## Hyd.Pressure2     Filler.Level      Filler.Speed      Temperature
## Min.      : -50.00      Min.      : 69.2      Min.      :1006      Min.      :63.80
## 1st Qu.: 0.00      1st Qu.:100.6      1st Qu.:3812      1st Qu.:65.40
## Median : 26.80      Median :118.6      Median :3978      Median :65.80
## Mean      : 20.11      Mean      :110.3      Mean      :3581      Mean      :66.23
## 3rd Qu.: 34.80      3rd Qu.:120.2      3rd Qu.:3996      3rd Qu.:66.60
## Max.      : 61.40      Max.      :153.2      Max.      :4020      Max.      :75.40
## NA's      :1          NA's      :2          NA's      :10      NA's      :2
## Usage.cont        Carb.Flow      PH              Oxygen.Filler
## Min.      :12.90      Min.      : 0      Mode:logical      Min.      :0.00240
## 1st Qu.:18.12      1st Qu.:1083      NA's:267          1st Qu.:0.01960
## Median :21.44      Median :3038          Median :0.03370
## Mean      :20.90      Mean      :2409          Mean      :0.04666
## 3rd Qu.:23.74      3rd Qu.:3215          3rd Qu.:0.05440
## Max.      :24.60      Max.      :3858          Max.      :0.39800
## NA's      :2          NA's      :3
## Air.Pressurer
## Min.      :141.2
## 1st Qu.:142.2
## Median :142.6
## Mean      :142.8
## 3rd Qu.:142.8
## Max.      :147.2
## NA's      :1
```

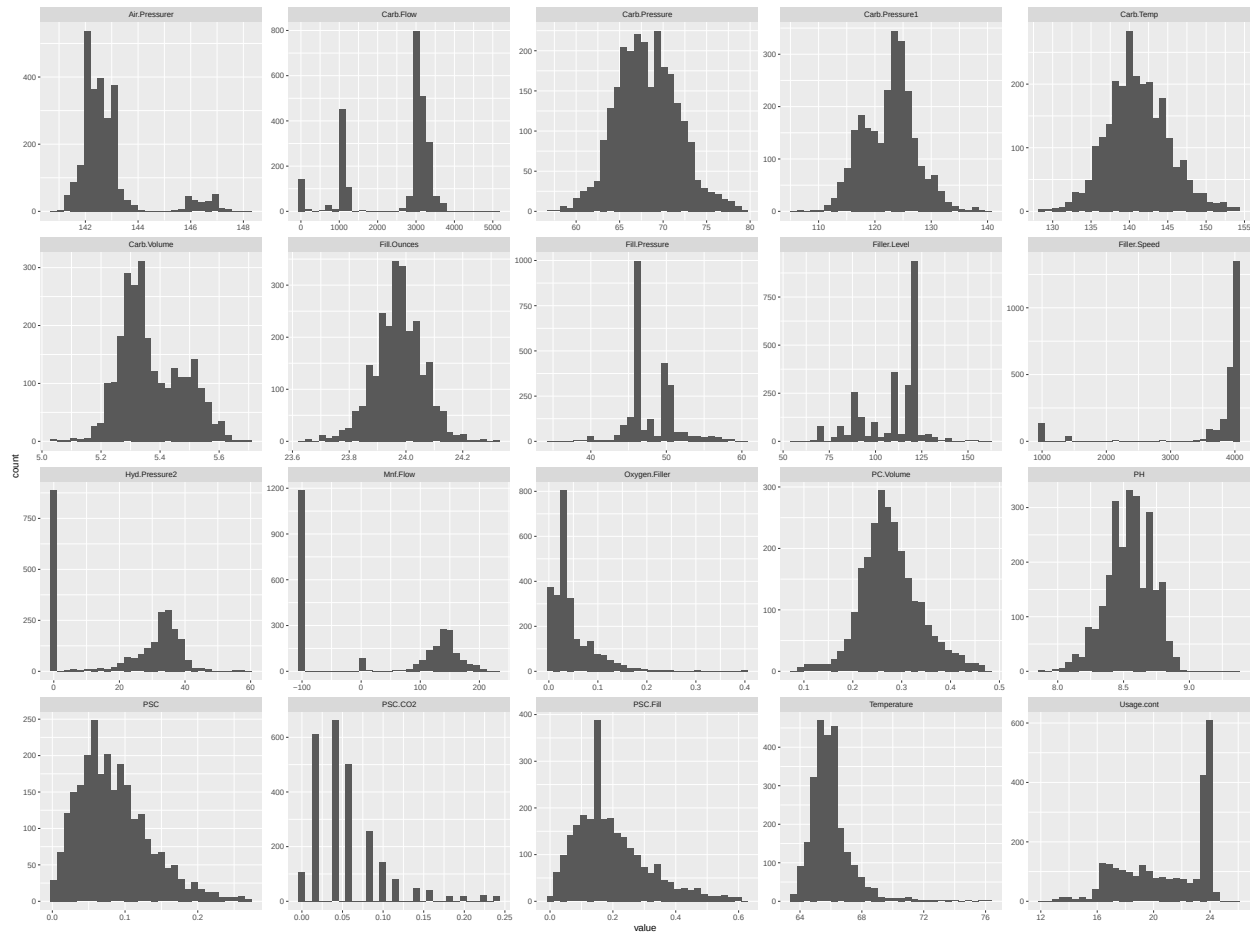
```
#there's also at least one really large value on oxygen filler here
summary(training_data_new)
```

```
## Brand.Code      Carb.Volume      Fill.Ounces      PC.Volume
## Length:2571     Min.    :5.040    Min.    :23.63    Min.    :0.07933
## Class :character 1st Qu.:5.293    1st Qu.:23.92    1st Qu.:0.23917
## Mode  :character Median :5.347    Median :23.97    Median :0.27133
##                Mean  :5.370    Mean  :23.97    Mean  :0.27712
##                3rd Qu.:5.453    3rd Qu.:24.03    3rd Qu.:0.31200
##                Max.  :5.700    Max.  :24.32    Max.  :0.47800
##                NA's  :10      NA's  :38      NA's  :39
## Carb.Pressure    Carb.Temp      PSC              PSC.Fill
## Min.    :57.00    Min.    :128.6    Min.    :0.00200    Min.    :0.0000
## 1st Qu.:65.60    1st Qu.:138.4    1st Qu.:0.04800    1st Qu.:0.1000
## Median :68.20    Median :140.8    Median :0.07600    Median :0.1800
## Mean    :68.19    Mean    :141.1    Mean    :0.08457    Mean    :0.1954
## 3rd Qu.:70.60    3rd Qu.:143.8    3rd Qu.:0.11200    3rd Qu.:0.2600
## Max.    :79.40    Max.    :154.0    Max.    :0.27000    Max.    :0.6200
## NA's    :27      NA's    :26      NA's    :33      NA's    :23
## PSC.CO2          Mnf.Flow      Carb.Pressure1    Fill.Pressure
## Min.    :0.00000    Min.    : -100.20    Min.    :105.6      Min.    :34.60
## 1st Qu.:0.02000    1st Qu.: -100.00    1st Qu.:119.0      1st Qu.:46.00
## Median :0.04000    Median : 65.20      Median :123.2      Median :46.40
## Mean    :0.05641    Mean    : 24.57      Mean    :122.6      Mean    :47.92
## 3rd Qu.:0.08000    3rd Qu.:140.80      3rd Qu.:125.4      3rd Qu.:50.00
## Max.    :0.24000    Max.    :229.40      Max.    :140.2      Max.    :60.40
## NA's    :39      NA's    :2        NA's    :32      NA's    :22
## Hyd.Pressure2    Filler.Level    Filler.Speed      Temperature      Usage.cont
## Min.    : 0.00    Min.    : 55.8      Min.    : 998      Min.    :63.60    Min.    :12.08
## 1st Qu.: 0.00    1st Qu.: 98.3      1st Qu.:3888      1st Qu.:65.20    1st Qu.:18.36
## Median :28.60    Median :118.4      Median :3982      Median :65.60    Median :21.79
## Mean    :20.96    Mean    :109.3      Mean    :3687      Mean    :65.97    Mean    :20.99
## 3rd Qu.:34.60    3rd Qu.:120.0      3rd Qu.:3998      3rd Qu.:66.40    3rd Qu.:23.75
## Max.    :59.40    Max.    :161.2      Max.    :4030      Max.    :76.20    Max.    :25.90
## NA's    :15      NA's    :20        NA's    :57      NA's    :14      NA's    :5
## Carb.Flow        PH          Oxygen.Filler      Air.Pressurer
## Min.    : 26      Min.    :7.880      Min.    :0.00240    Min.    :140.8
## 1st Qu.:1144      1st Qu.:8.440      1st Qu.:0.02200    1st Qu.:142.2
## Median :3028      Median :8.540      Median :0.03340    Median :142.6
## Mean    :2468      Mean    :8.546      Mean    :0.04684    Mean    :142.8
## 3rd Qu.:3186      3rd Qu.:8.680      3rd Qu.:0.06000    3rd Qu.:143.0
## Max.    :5104      Max.    :9.360      Max.    :0.40000    Max.    :148.2
## NA's    :2        NA's    :4        NA's    :12
```

After quickly browsing the remaining variables, while some of the fields have a wide range of values, none of the values look like errors (there are clusters of similar low/high values for each variable). While they may count as outliers (we'll calculate later) they are likely legitimate data and not input errors.

Let's check histograms to see if anything stands out:

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



```
## Brand.Code    n
## 1            A 293
## 2            B 1239
## 3            C 304
## 4            D 615
## 5          <NA> 120
```

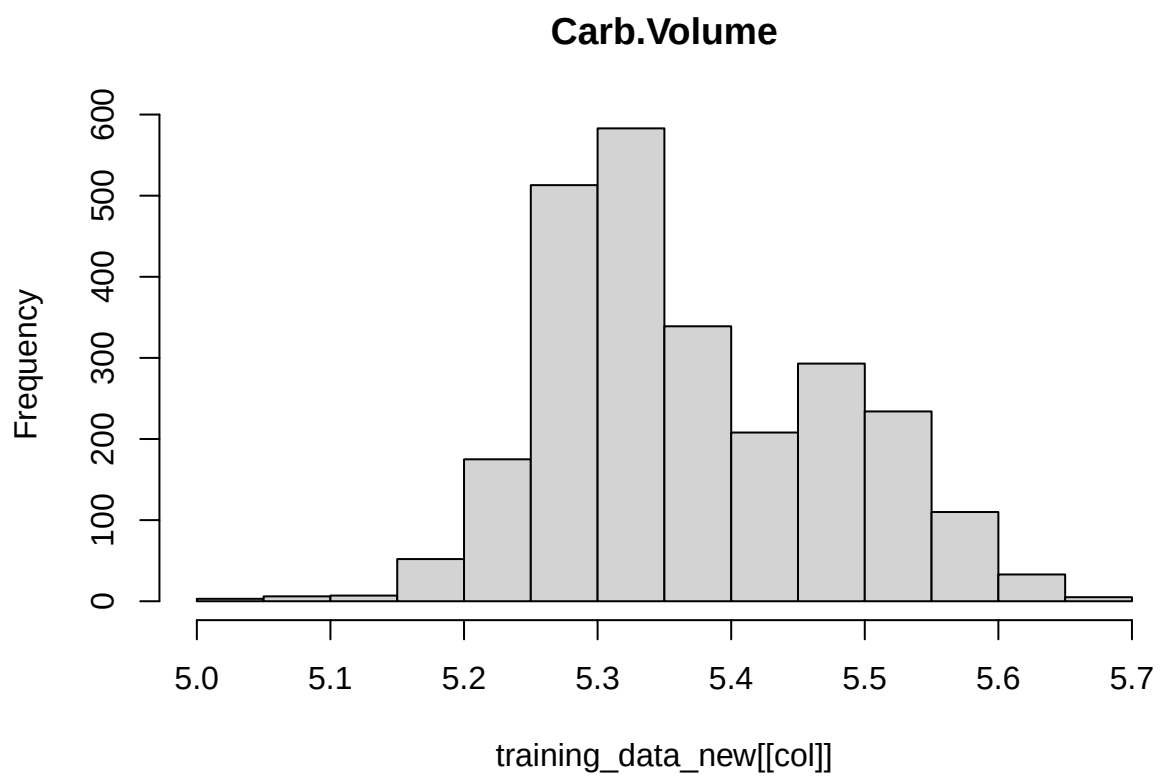
Not all the data is normally distributed. pH, the response variable is normally distributed. Other variables close to a normal distribution include carb pressure, carb pressure 1 (slightly bimodal), carb temp, fill ounces, carb volume (also slightly bimodal), psc (skewed slightly right), pc volume, temperature (skewed slightly right), and psc fill (skewed right with a large peak).

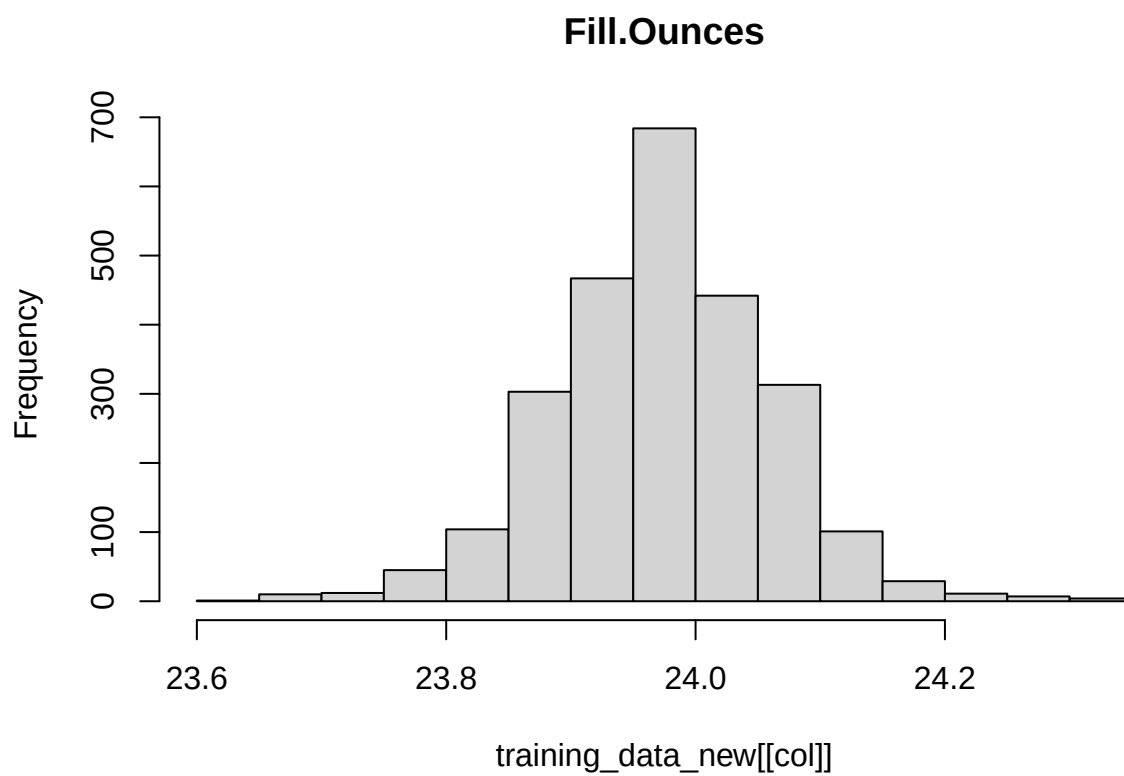
A few are more clearly bimodal/have very different peaks on either side of the graph: air pressurer, carb flow, fill pressure, filler level, filler speed, , hyd pressure 2, mnf flow, and usage cont.

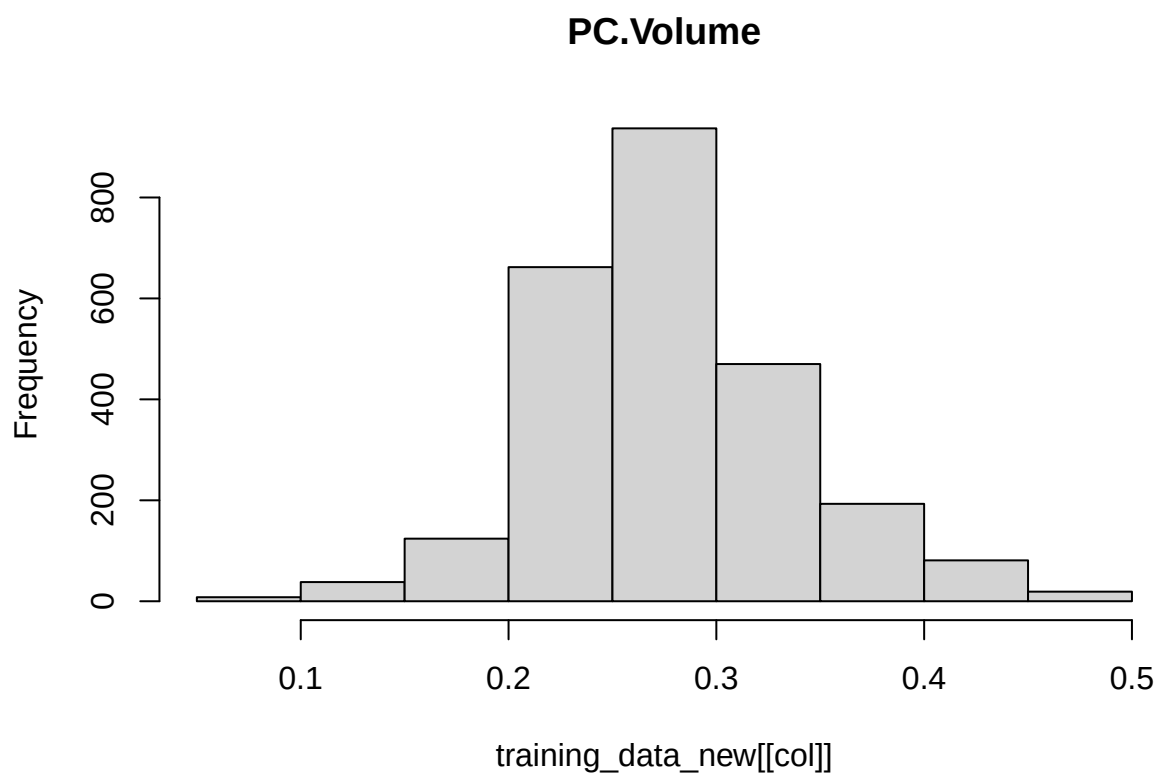
Finally, some are skewed right: in addition to the variables mentioned above, oxygen filler looks like it may have some outliers, and psc co2 has a strange distribution, where the variable clusters around certain values, almost like a discrete variable.

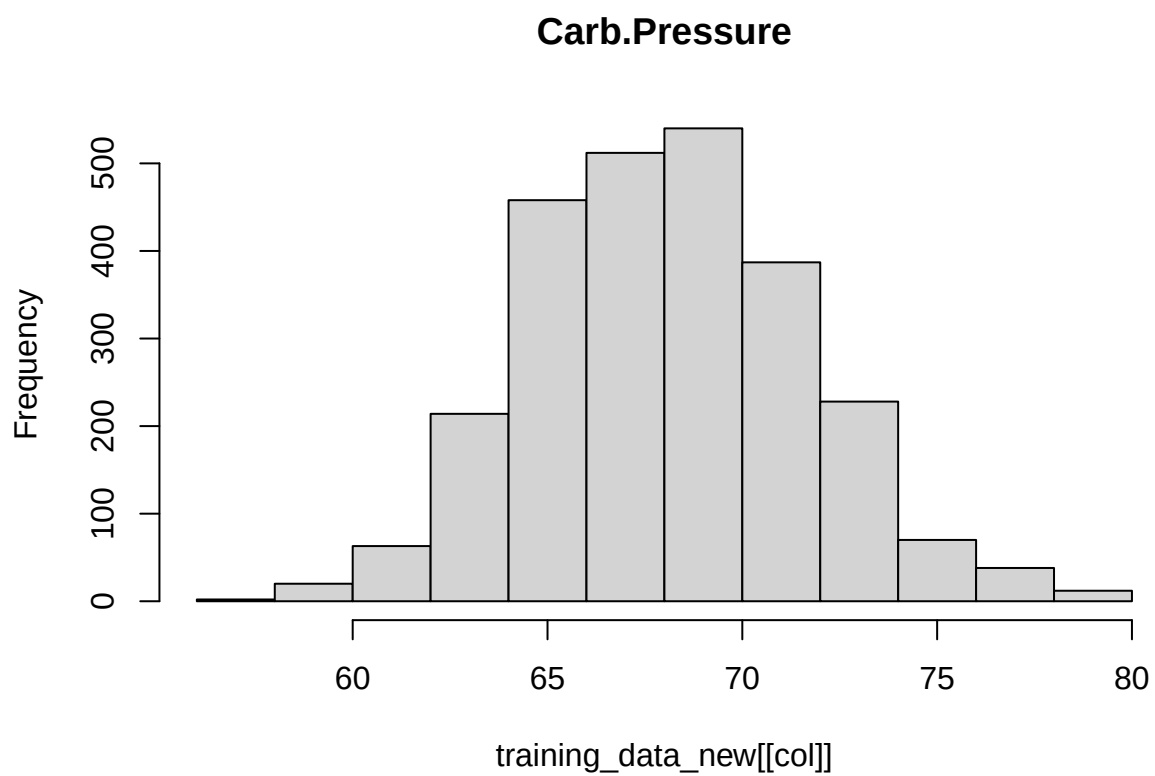
```
#histograms

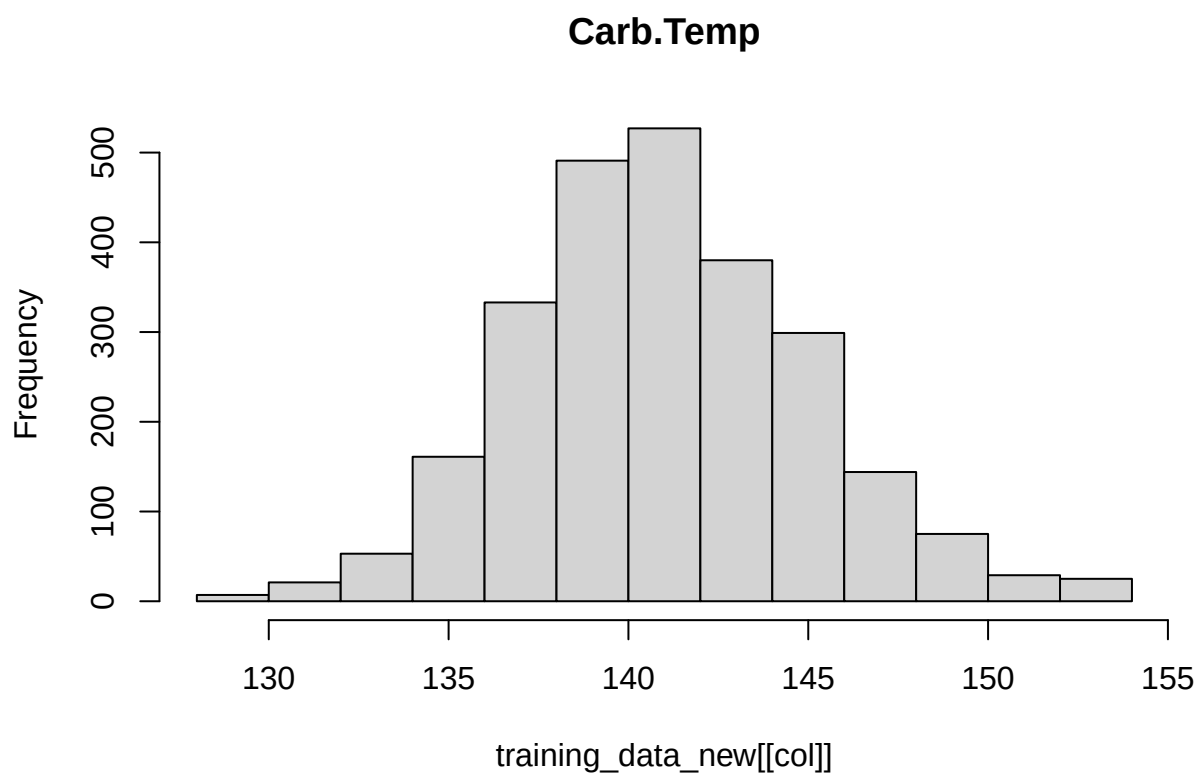
for (col in names(training_data_new %>% select(-Brand.Code))) {
  hist(training_data_new[[col]], main=col)
}
```



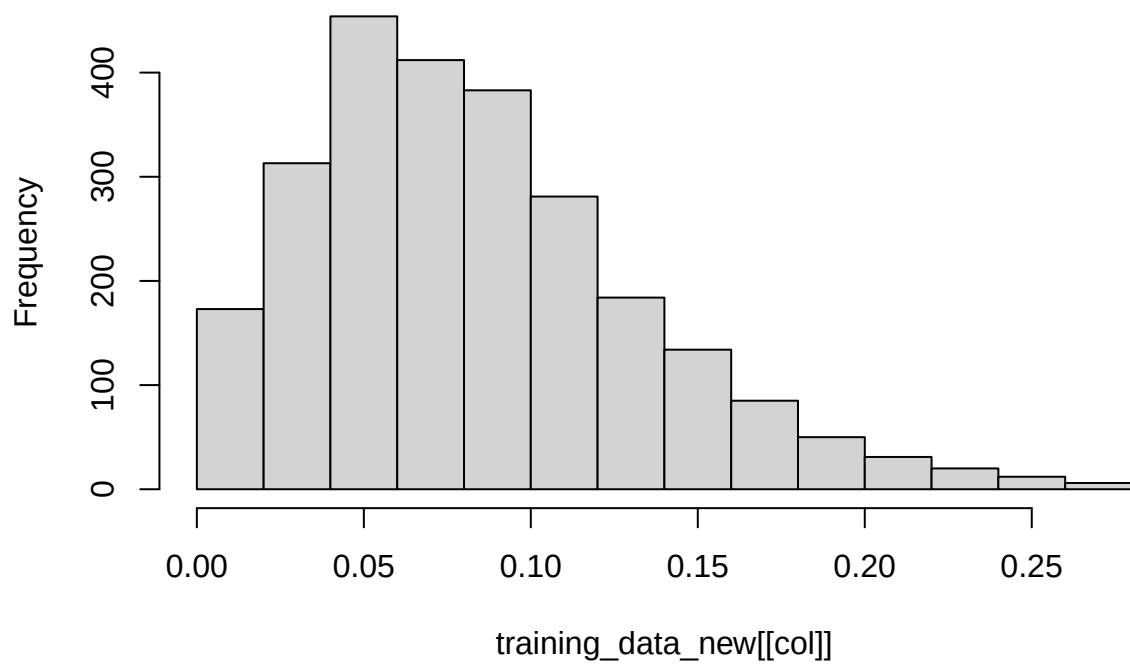




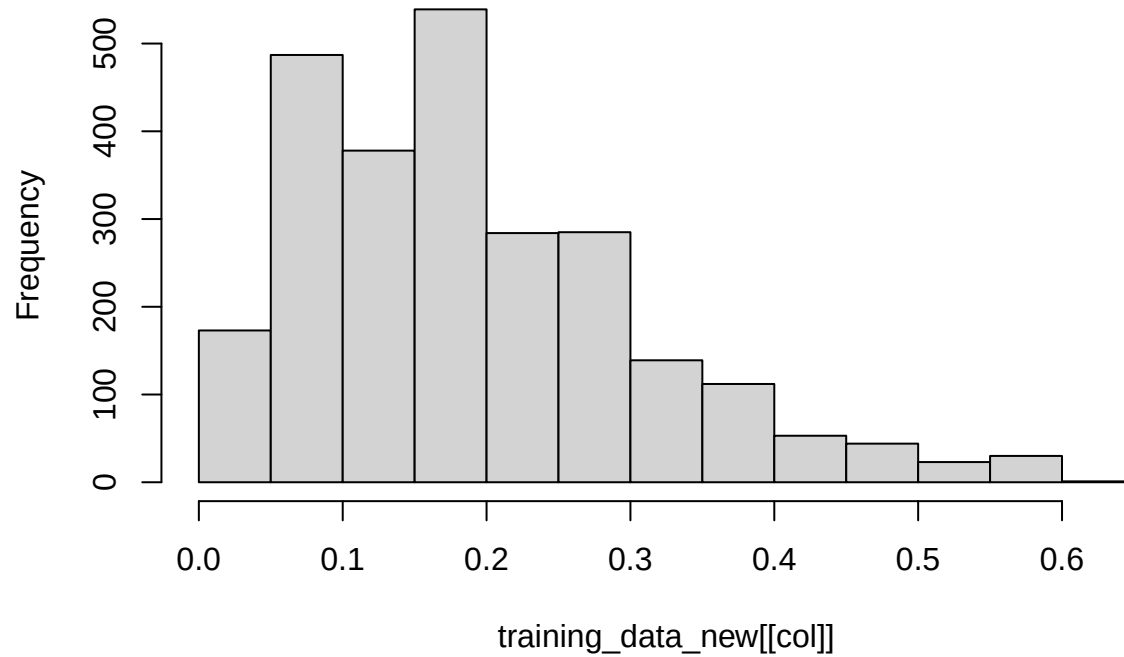




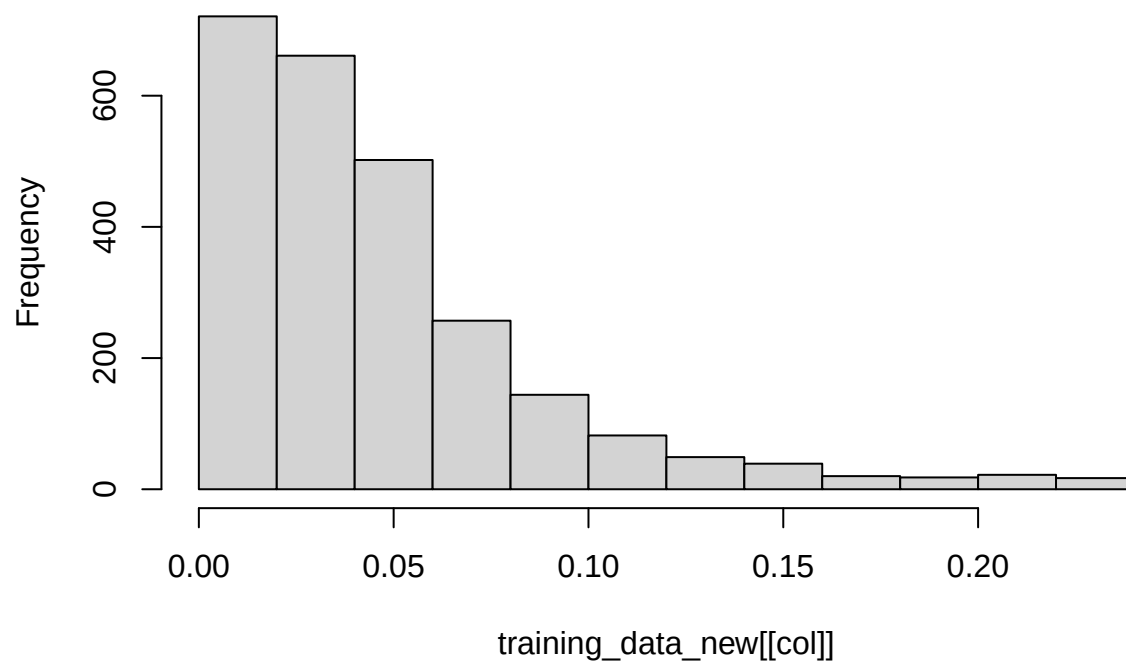
PSC



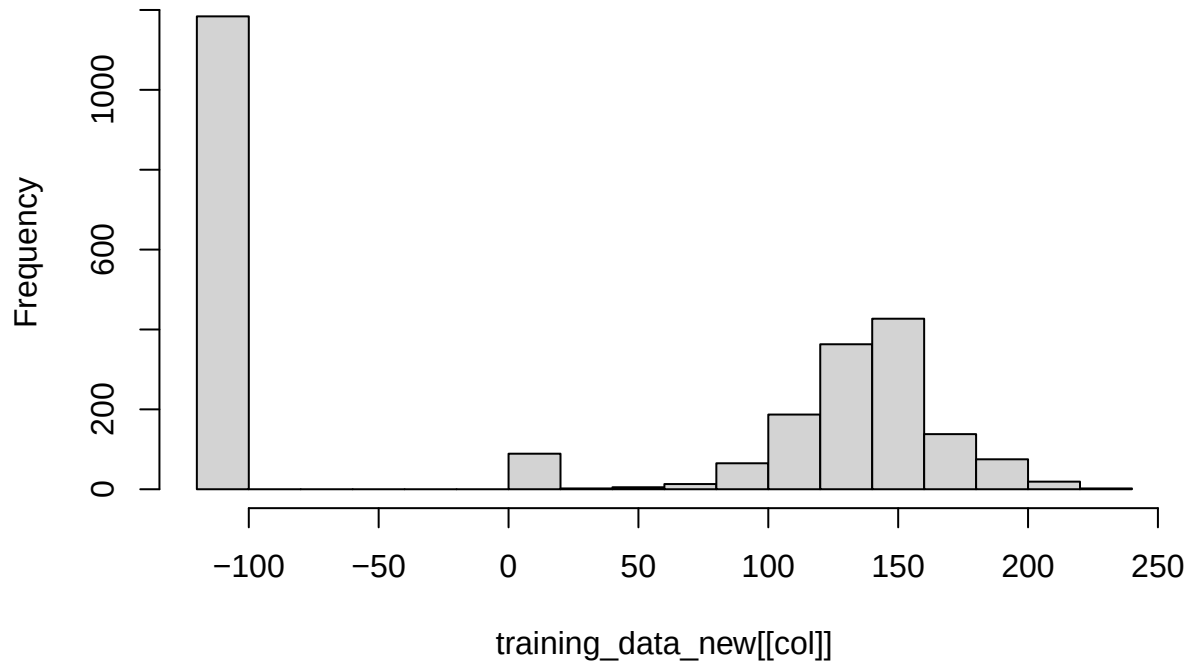
PSC.Fill

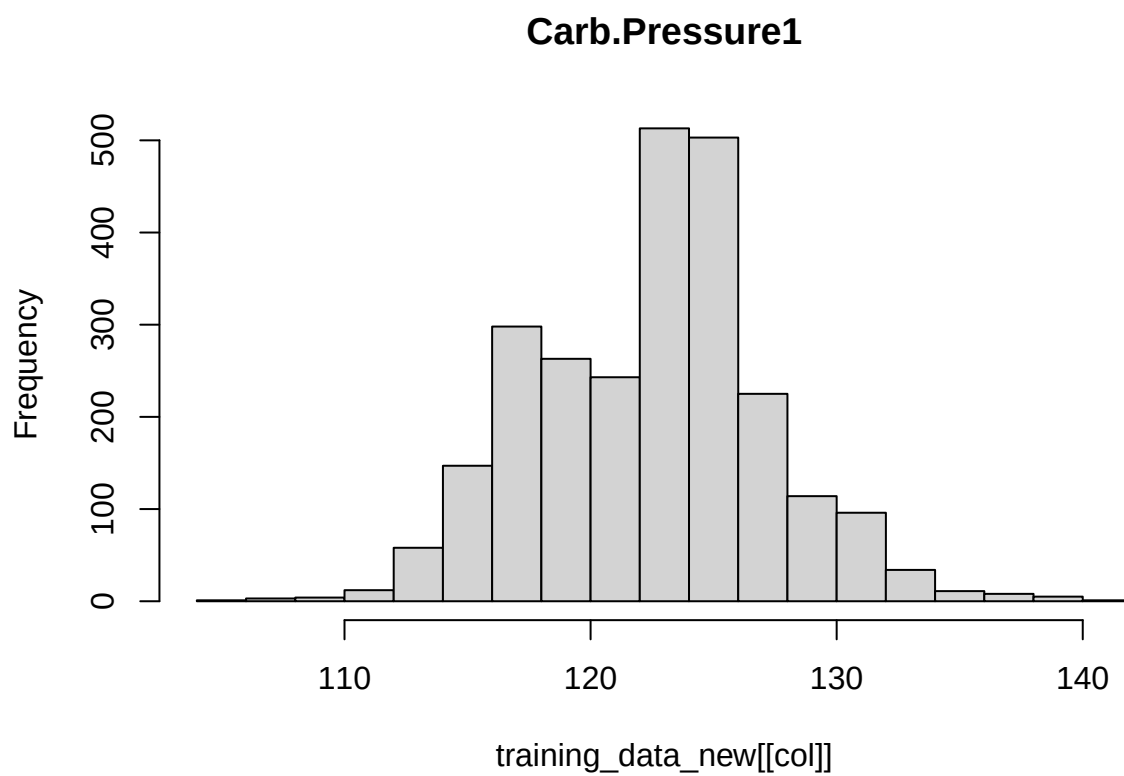


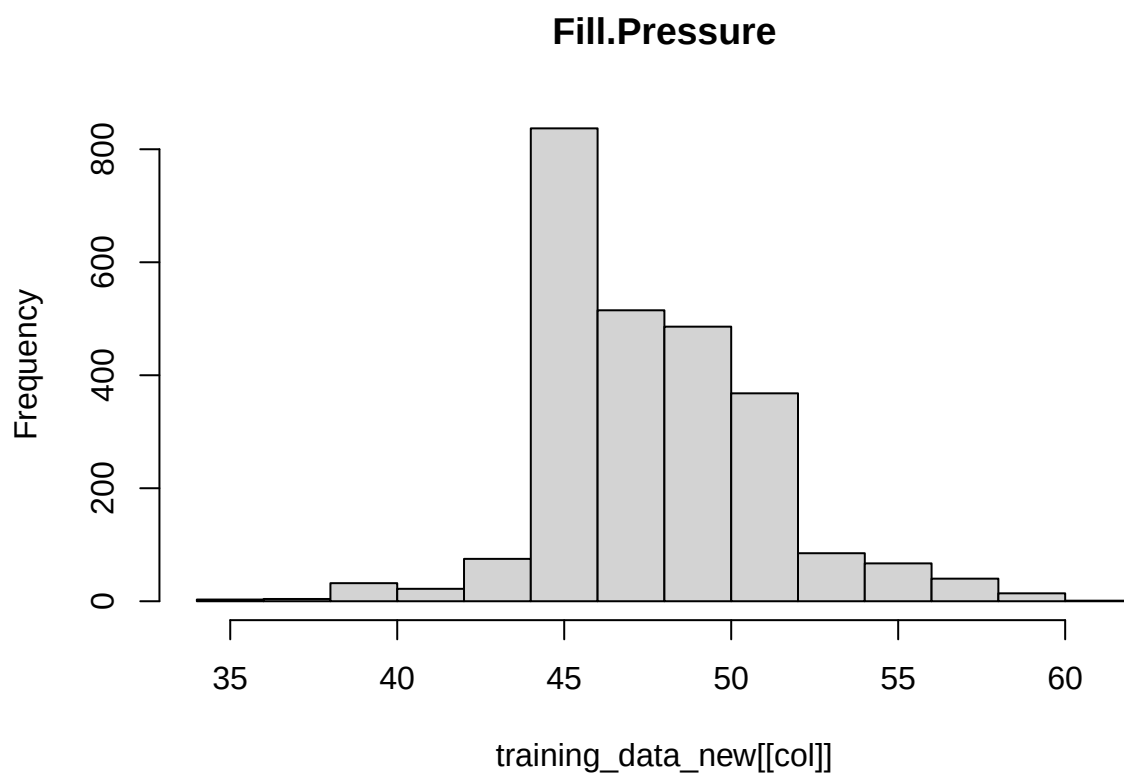
PSC.CO2



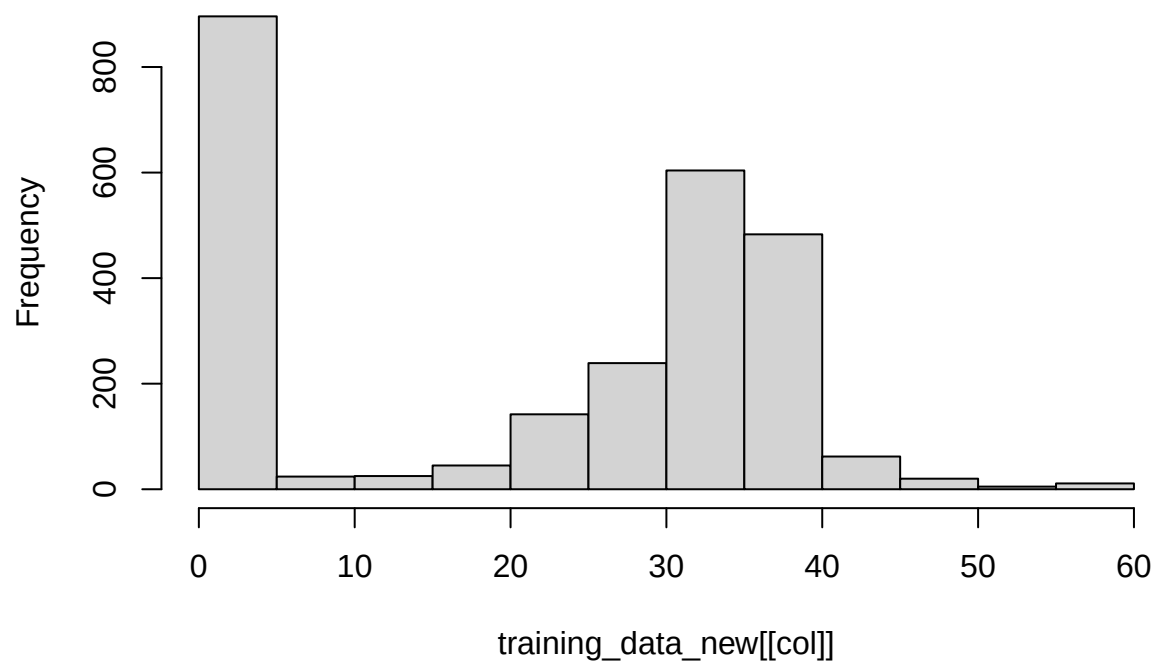
Mnf.Flow

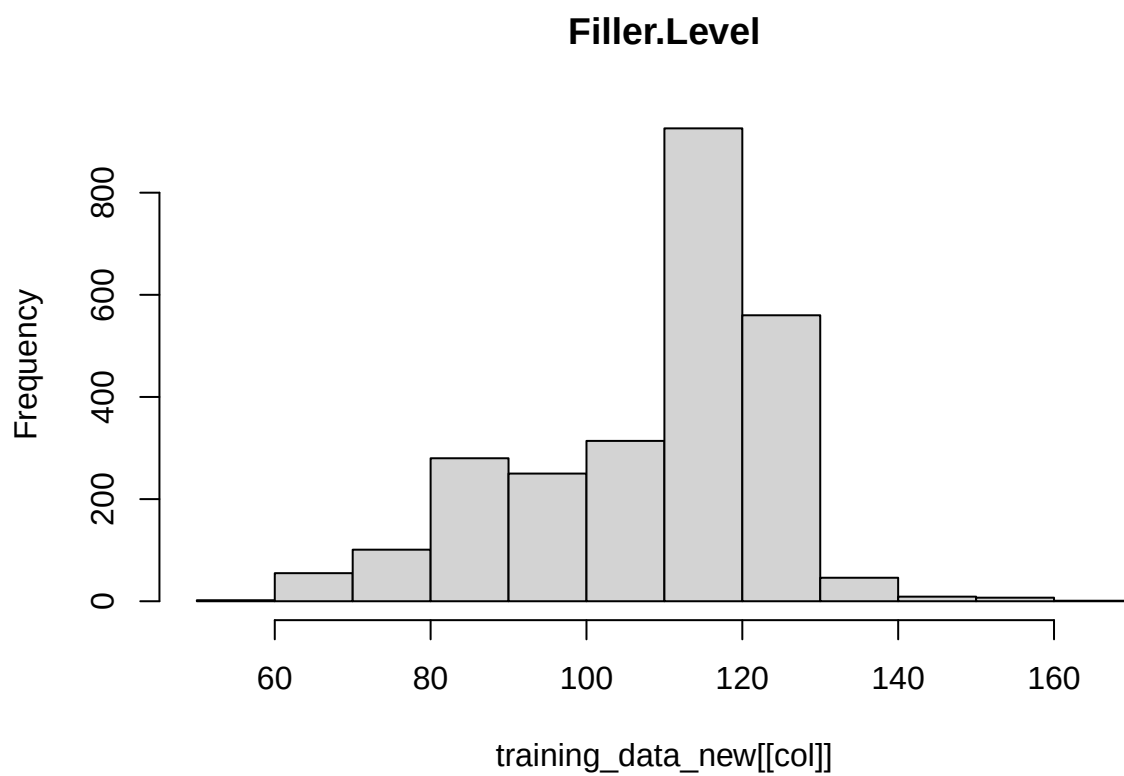


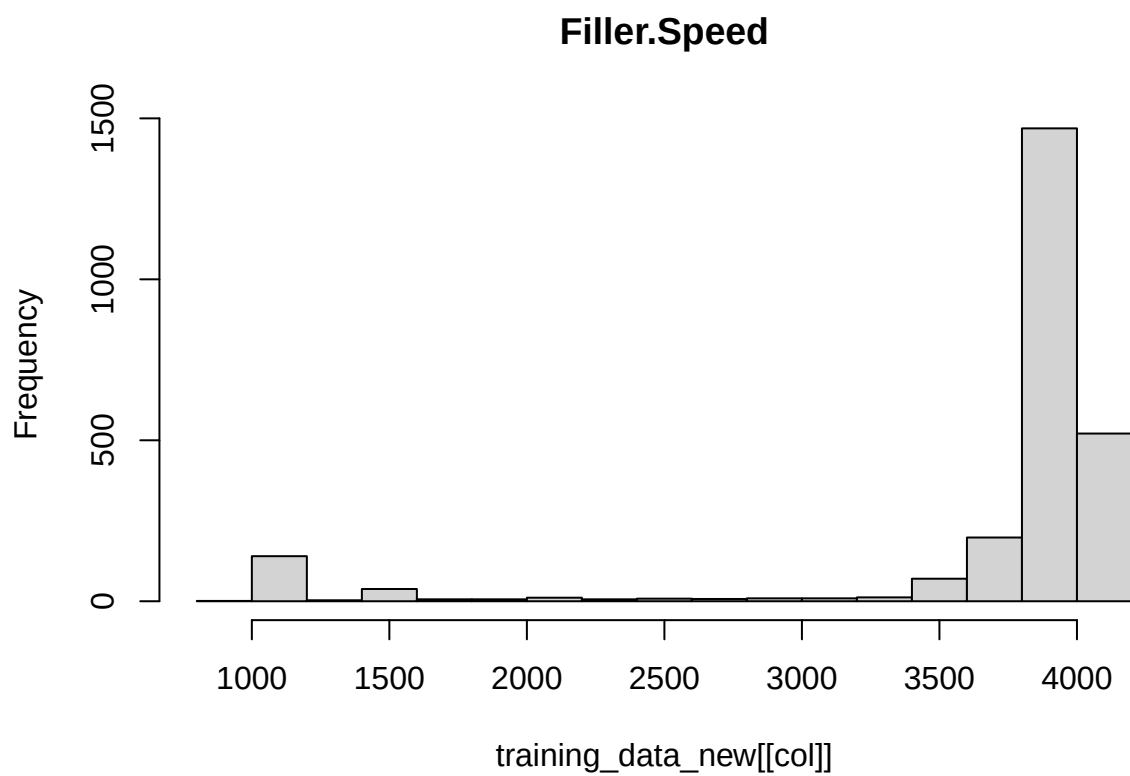


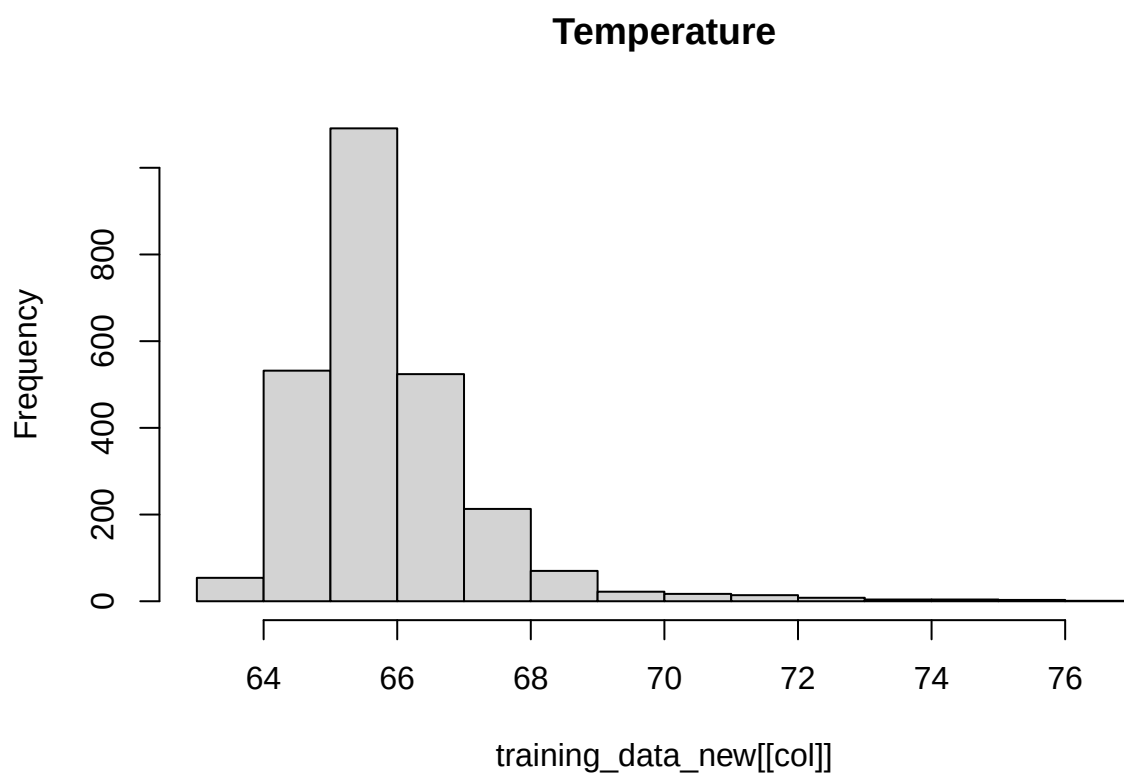


Hyd.Pressure2

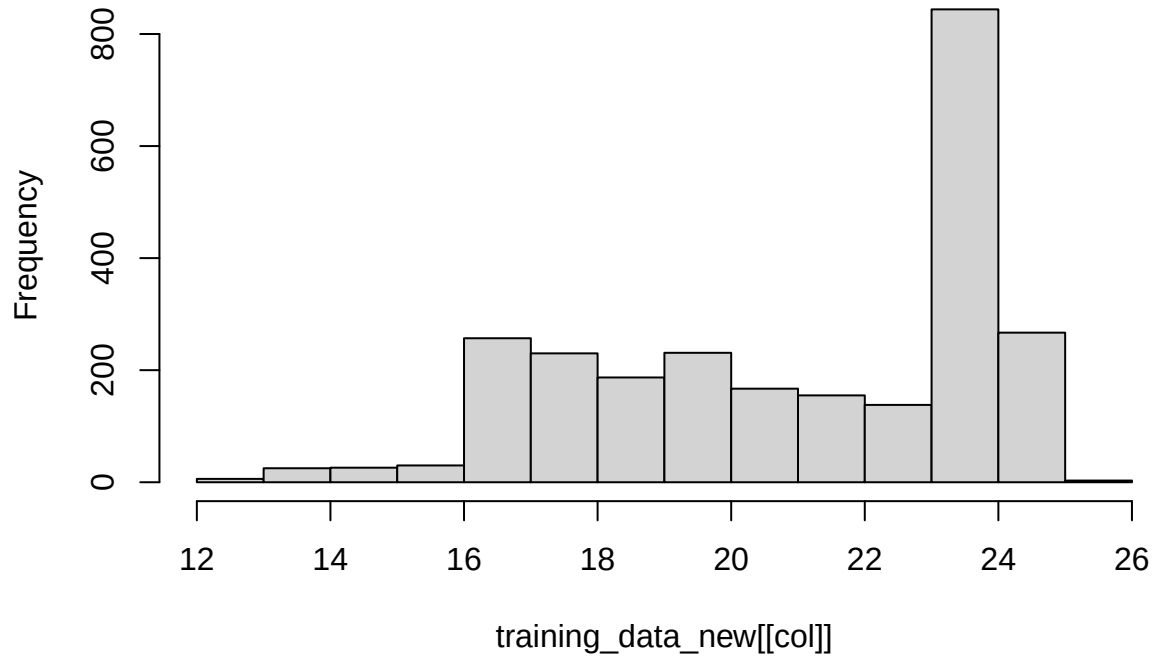


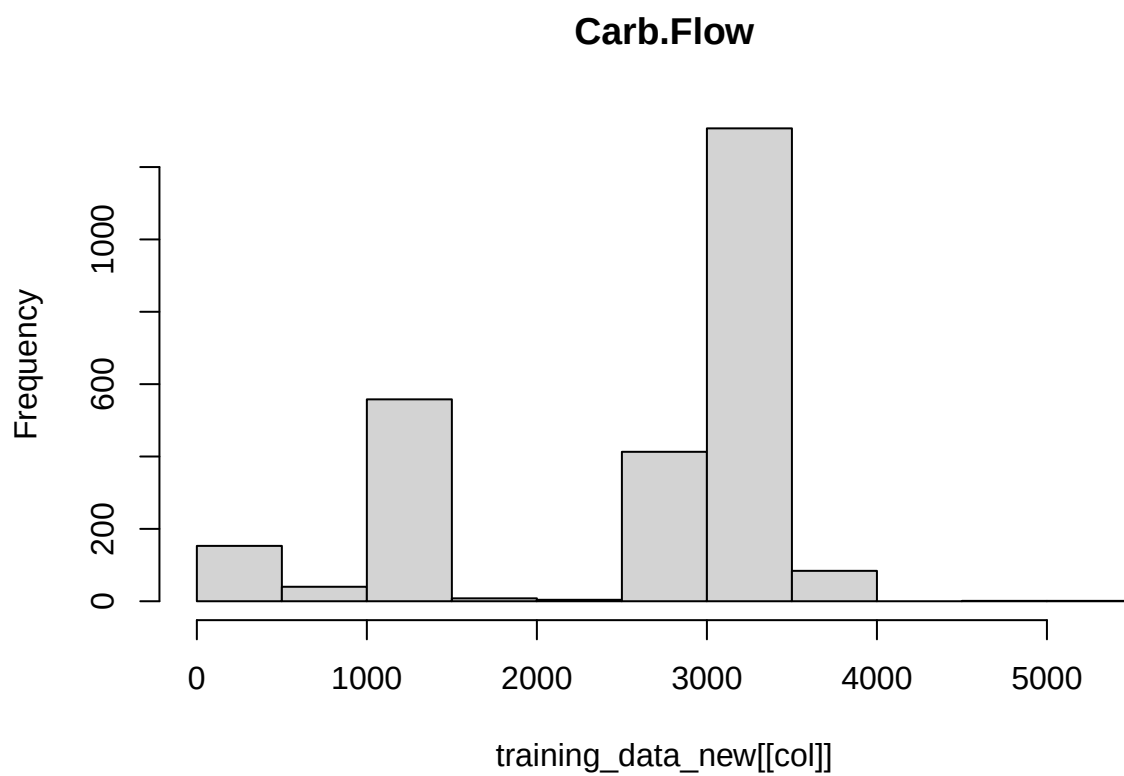


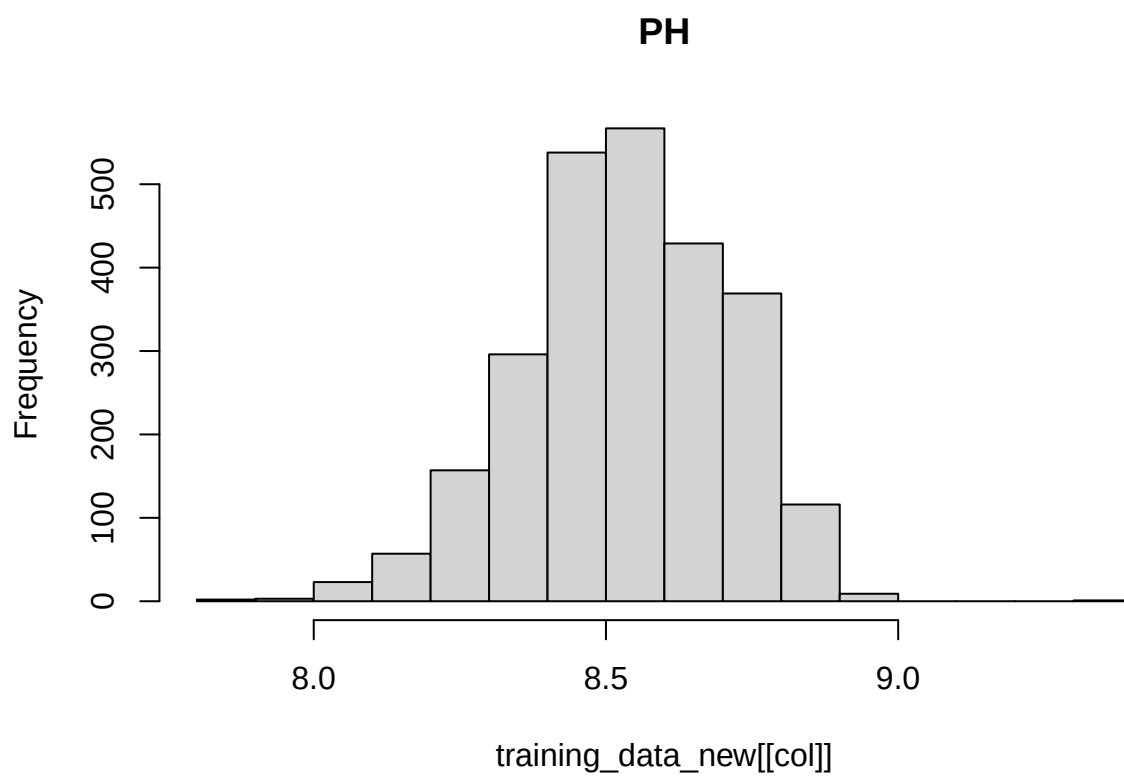




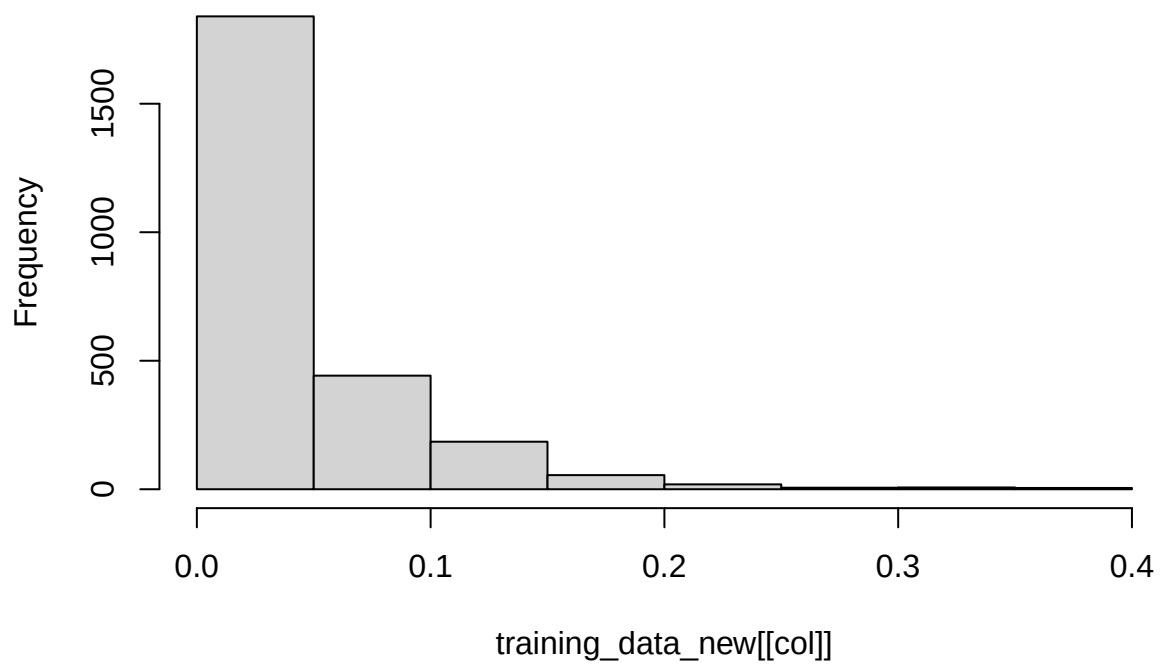
Usage.cont



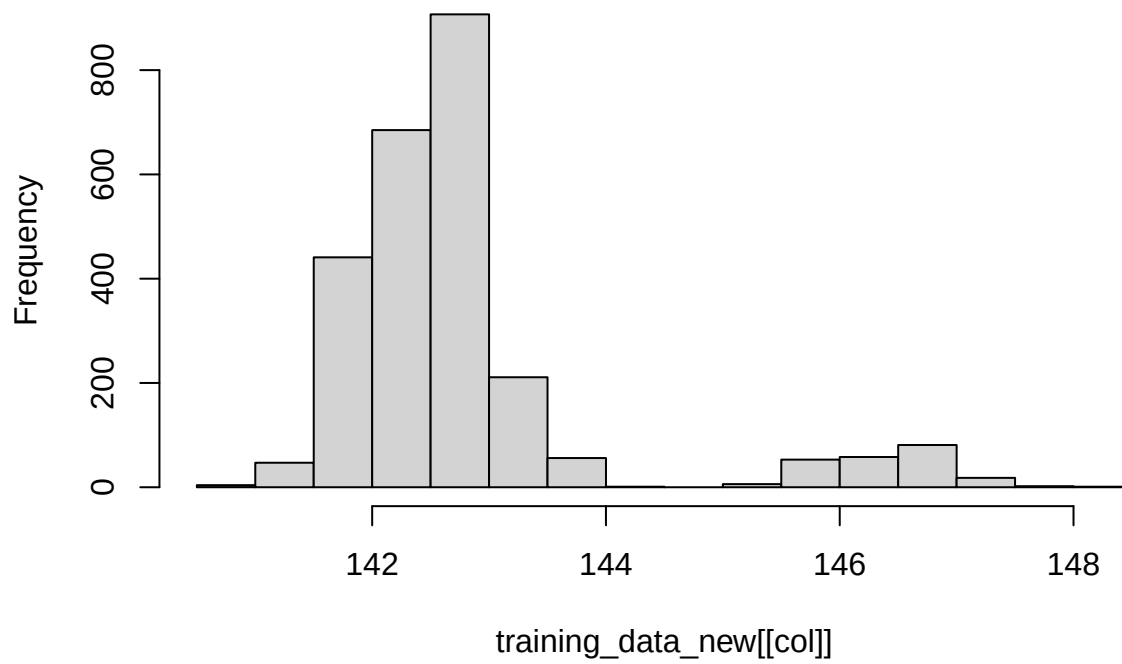




Oxygen.Filler

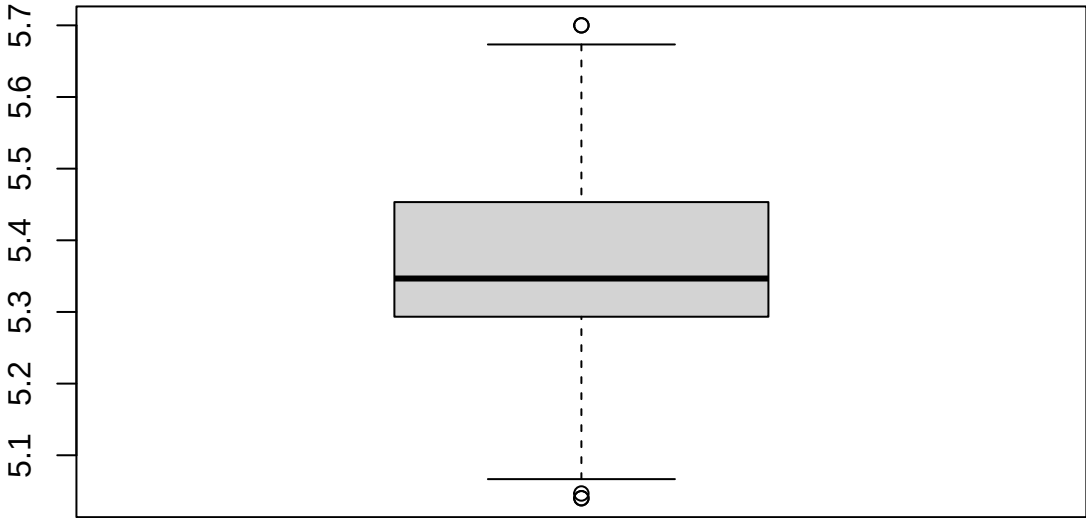


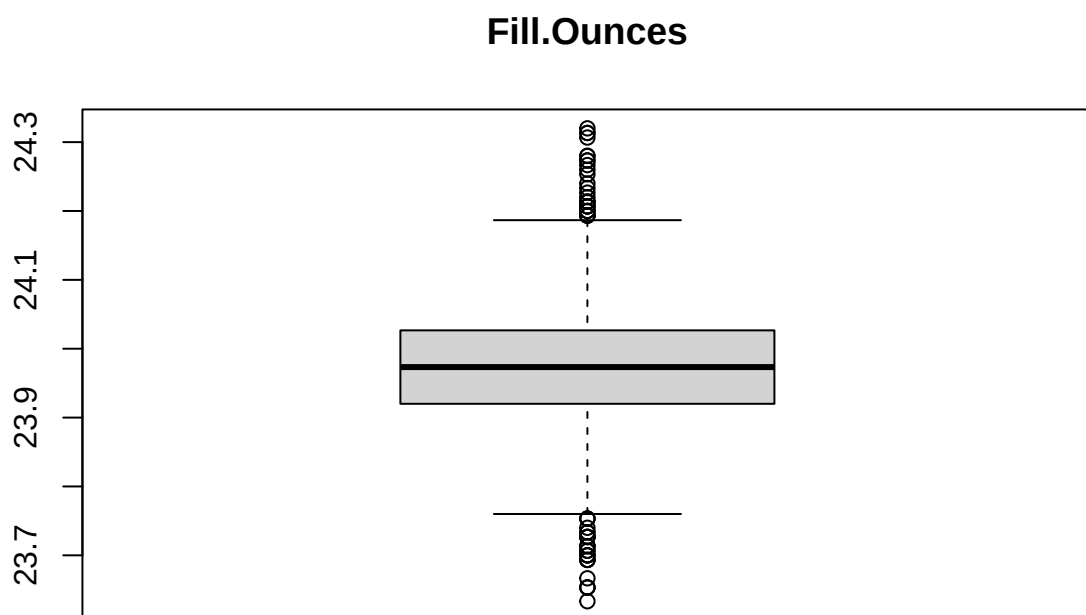
Air.Pressurer



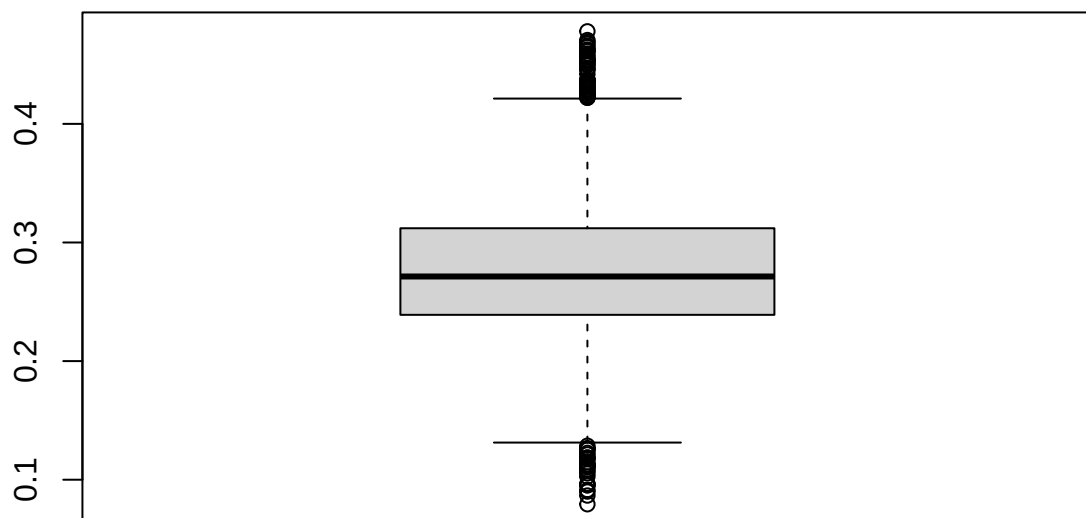
```
#box plots
for (col in names(training_data_new %>% select(-Brand.Code))) {
  boxplot(training_data_new[[col]], main=col)
}
```

Carb.Volume

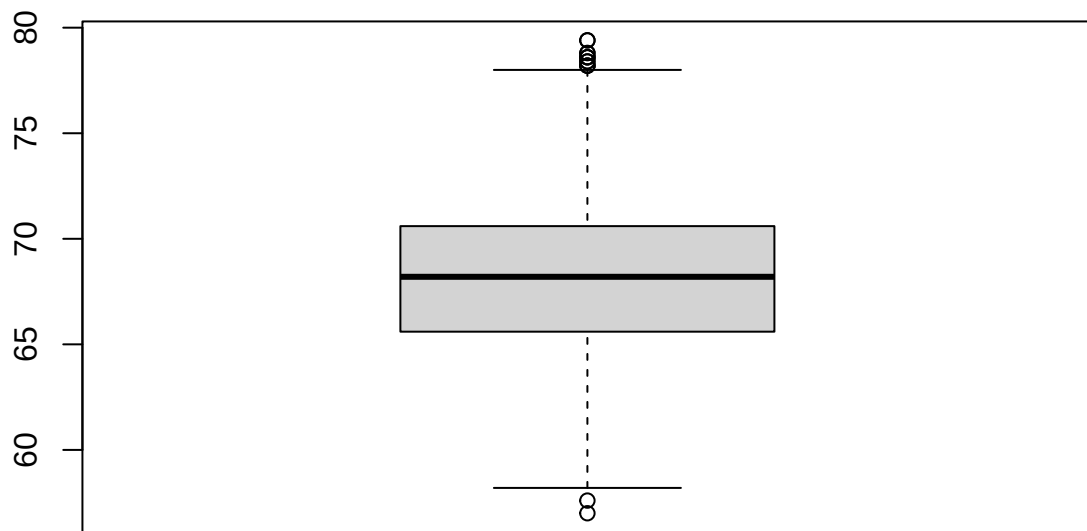


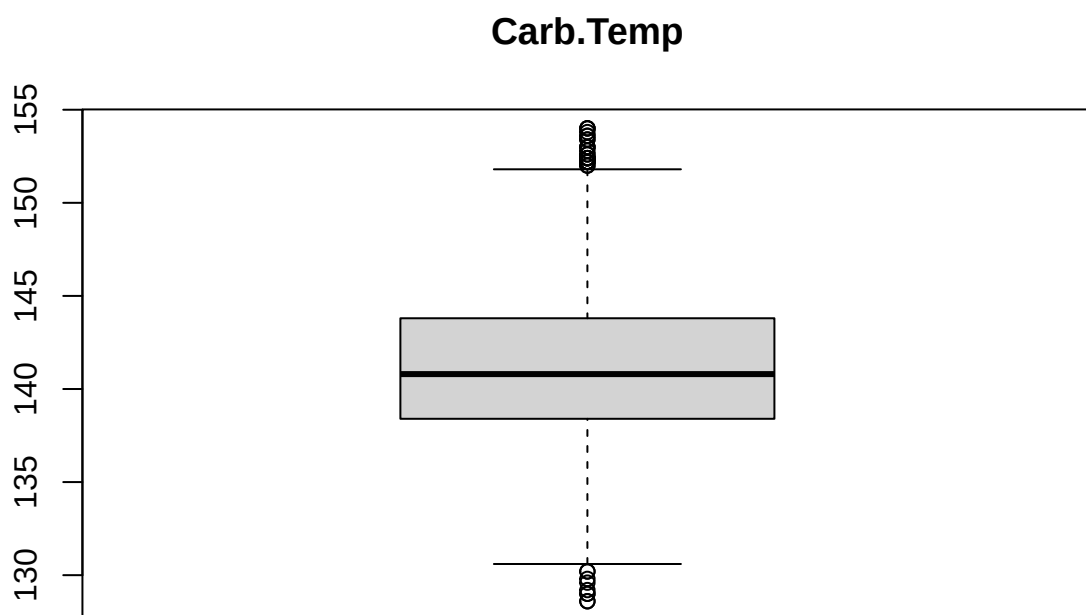


PC.Volume

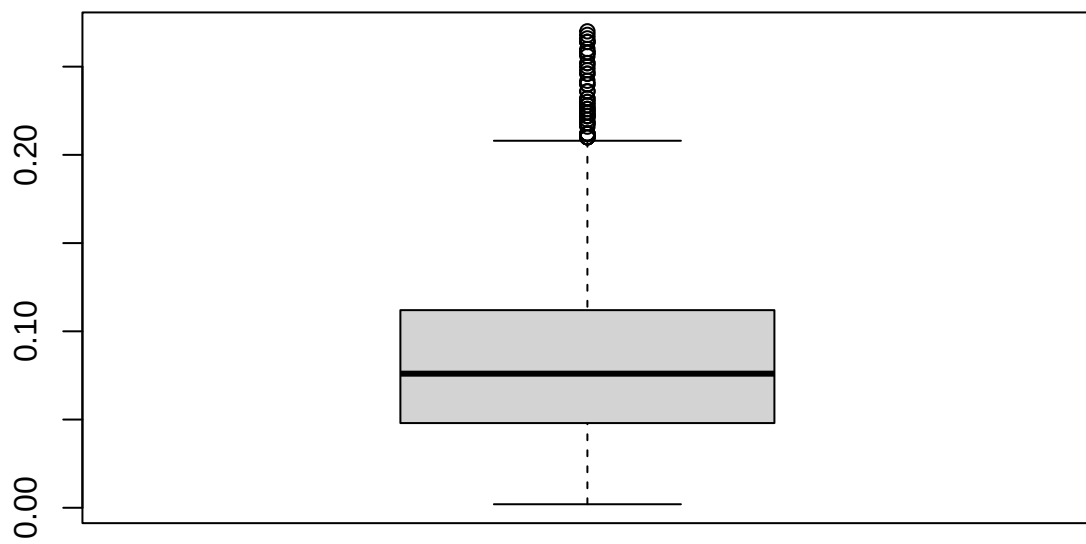


Carb.Pressure

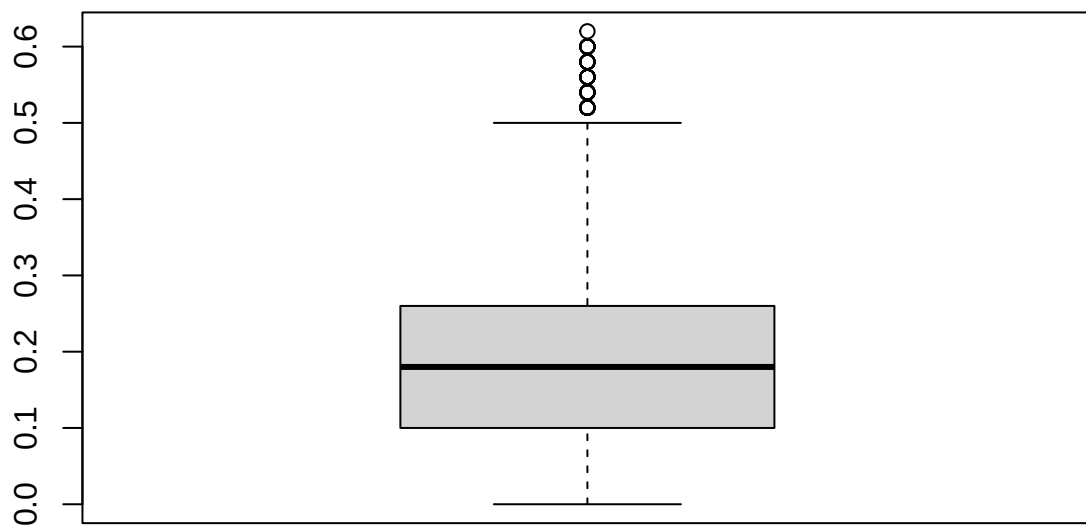




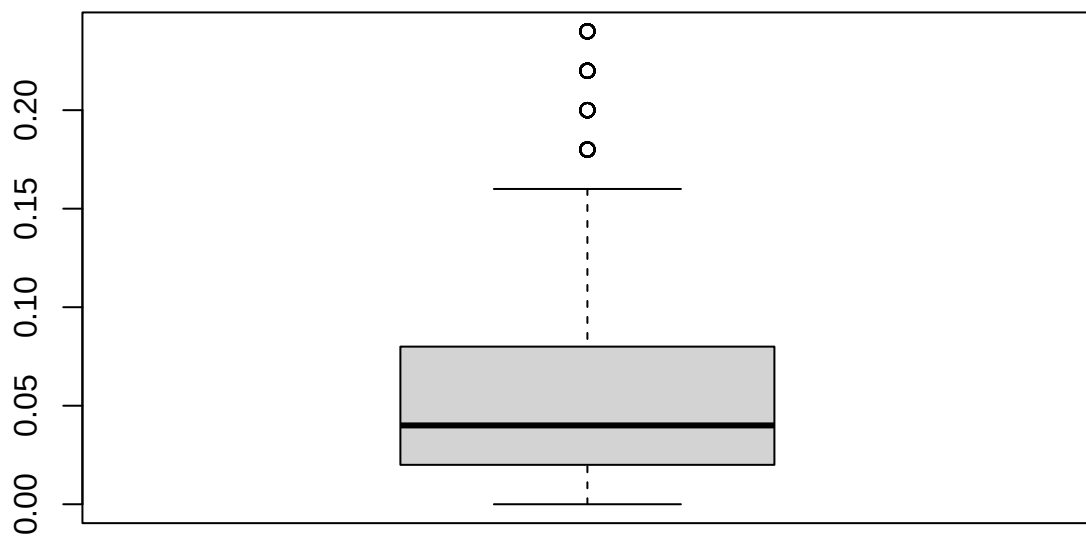
PSC



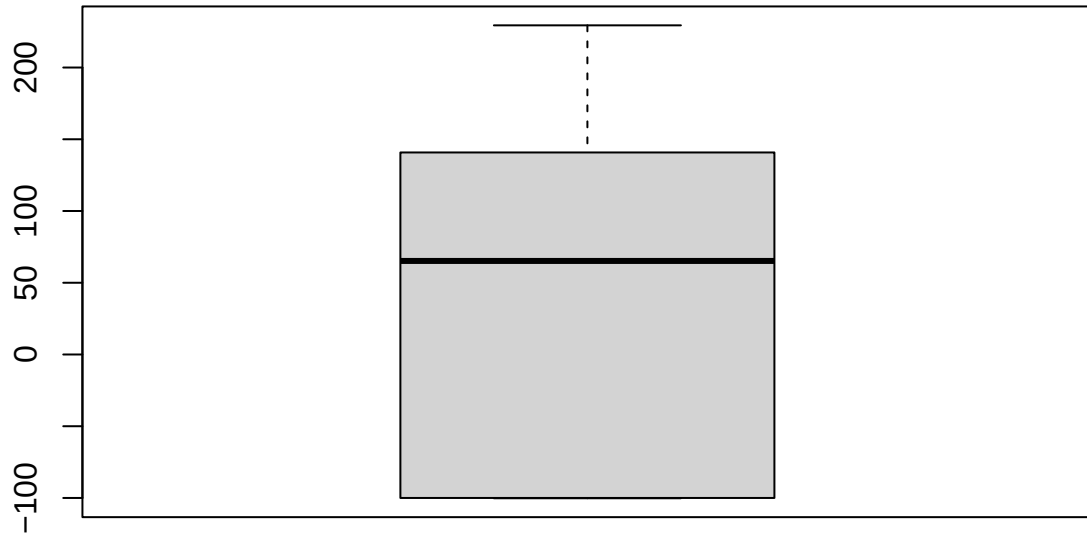
PSC.Fill



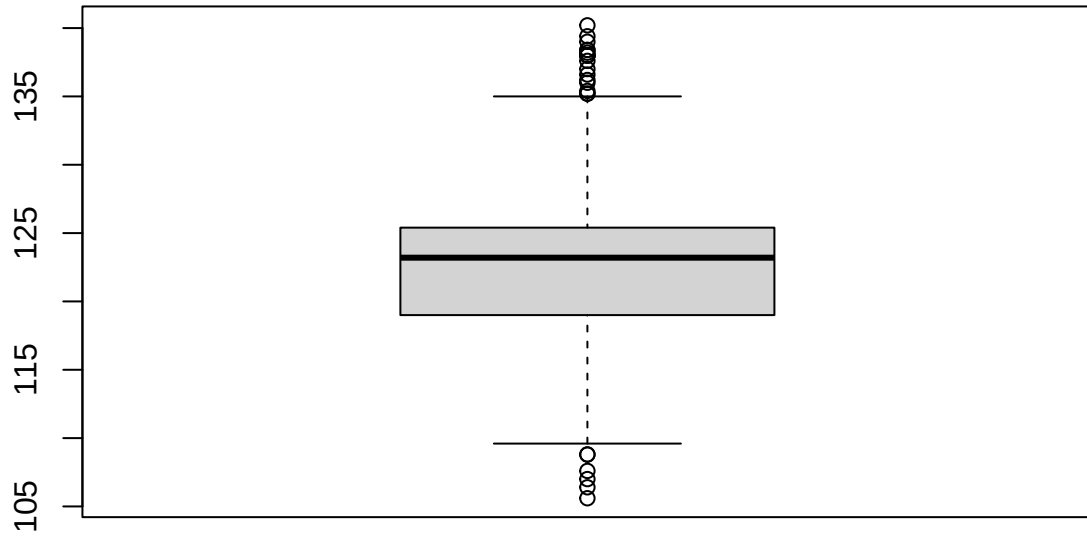
PSC.CO2



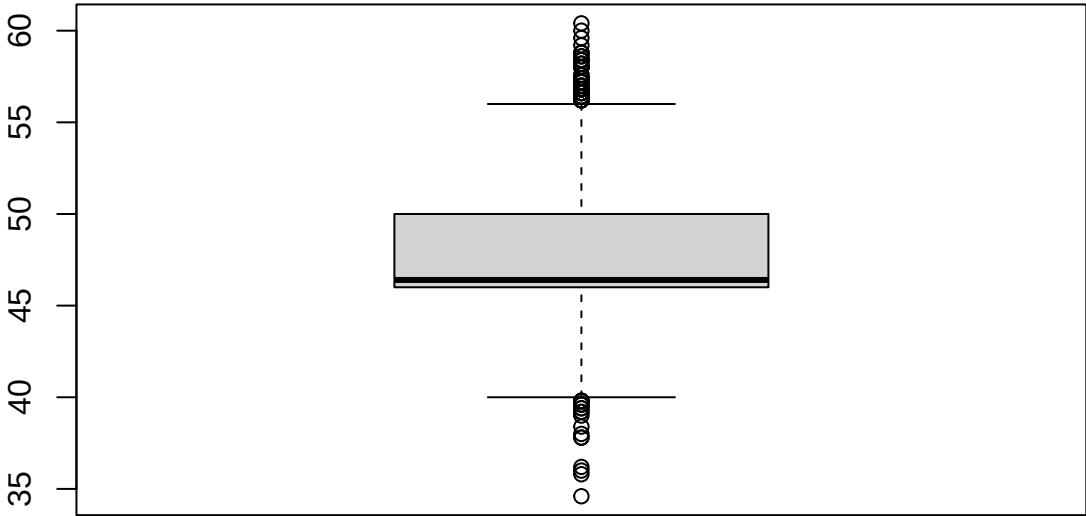
Mnf.Flow



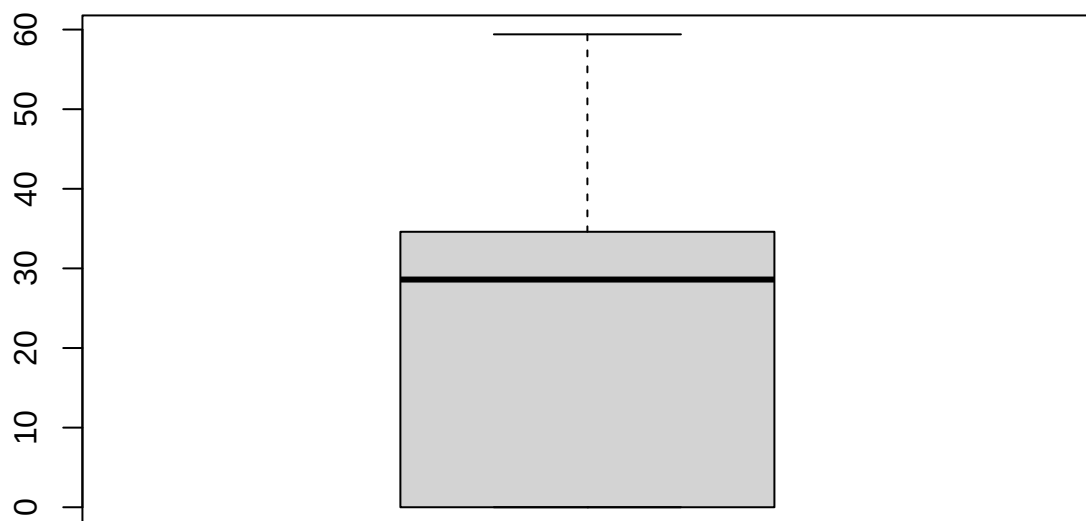
Carb.Pressure1

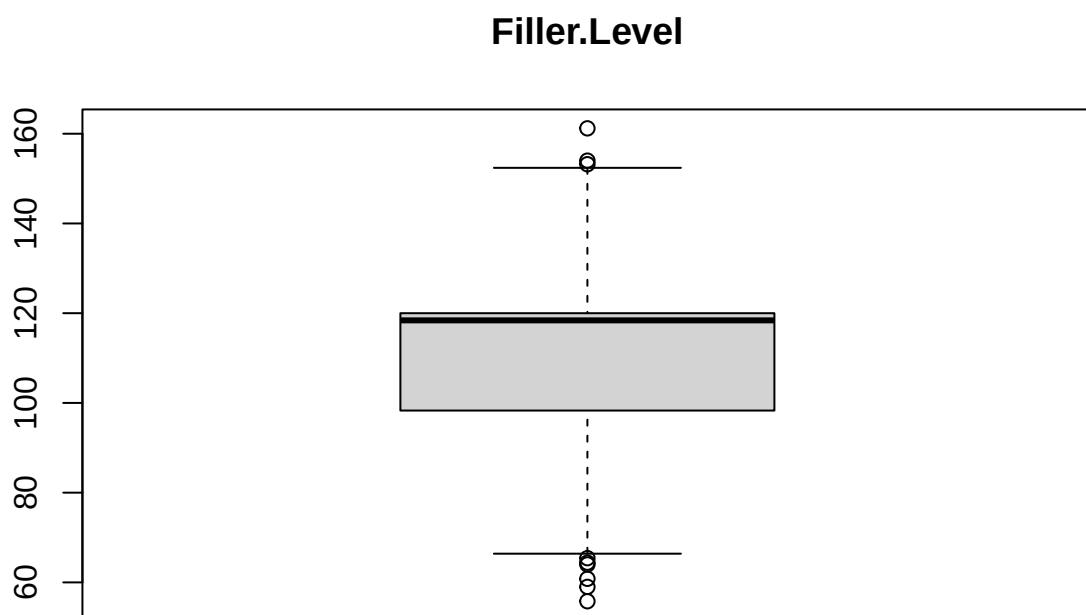


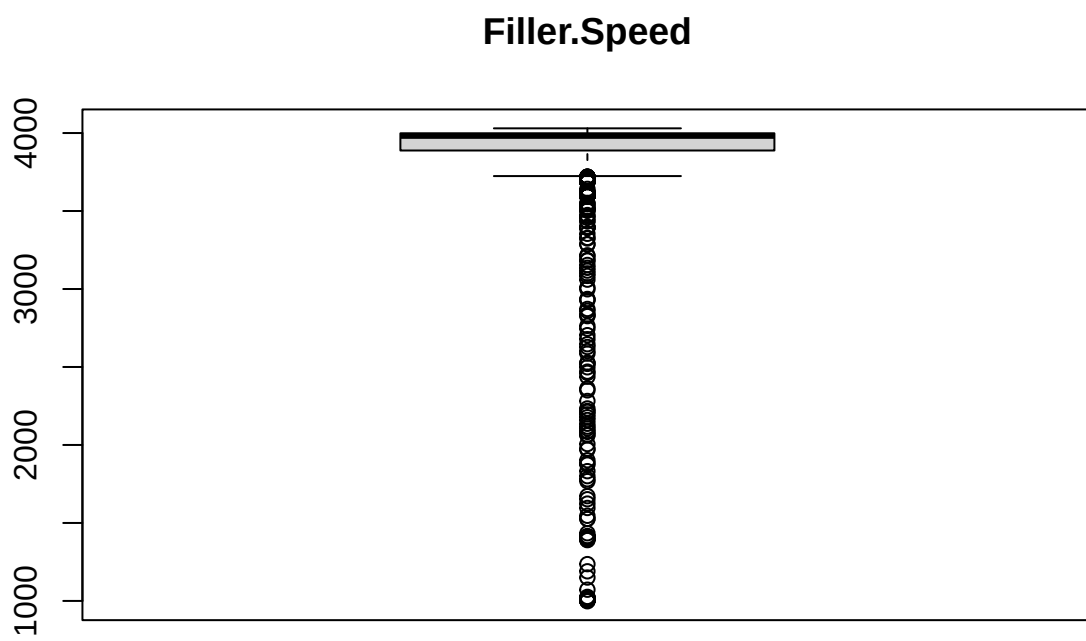
Fill.Pressure



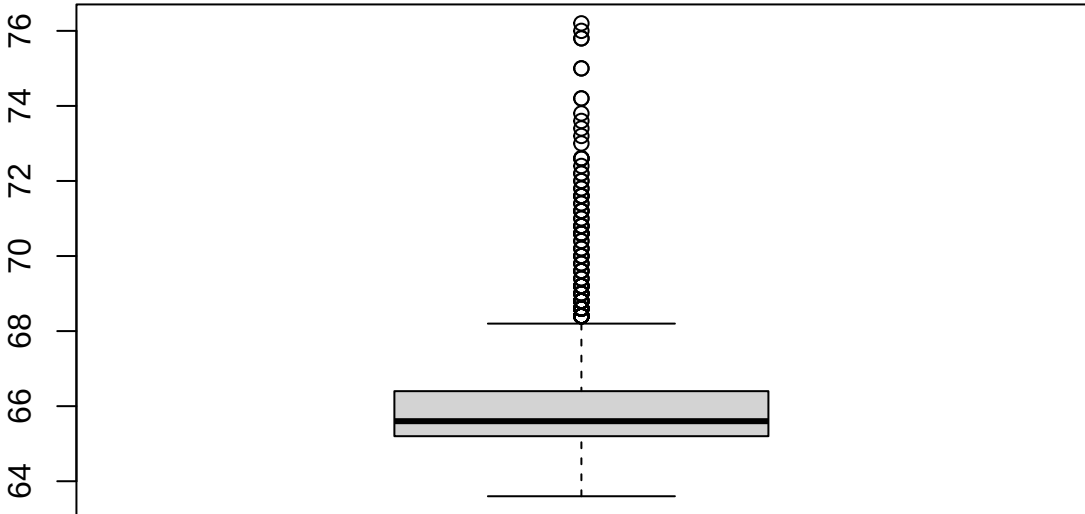
Hyd.Pressure2



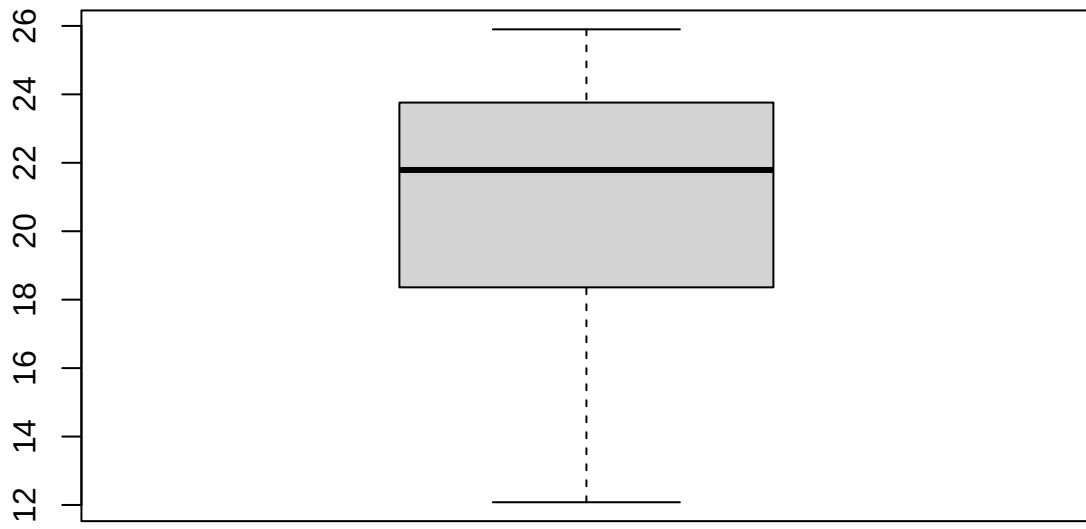




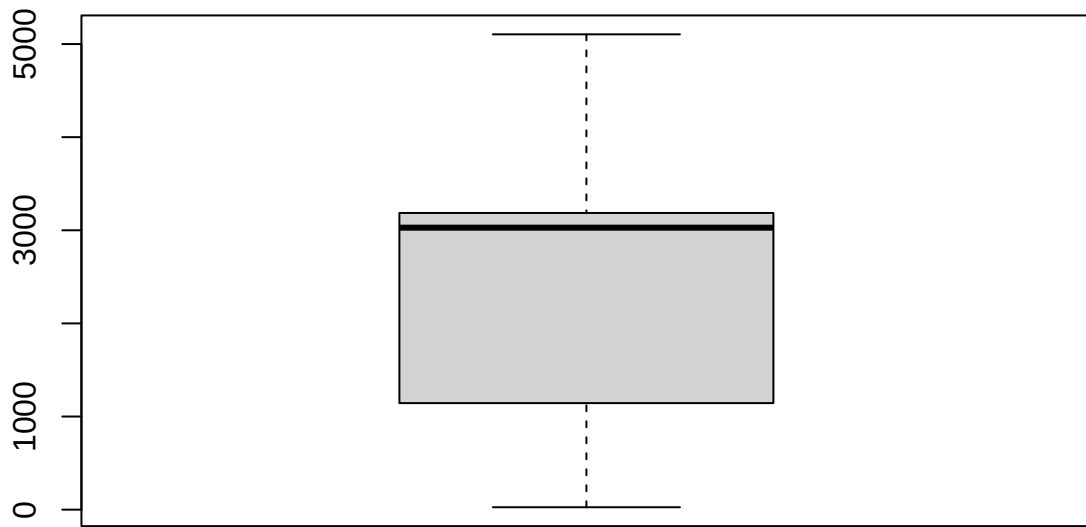
Temperature

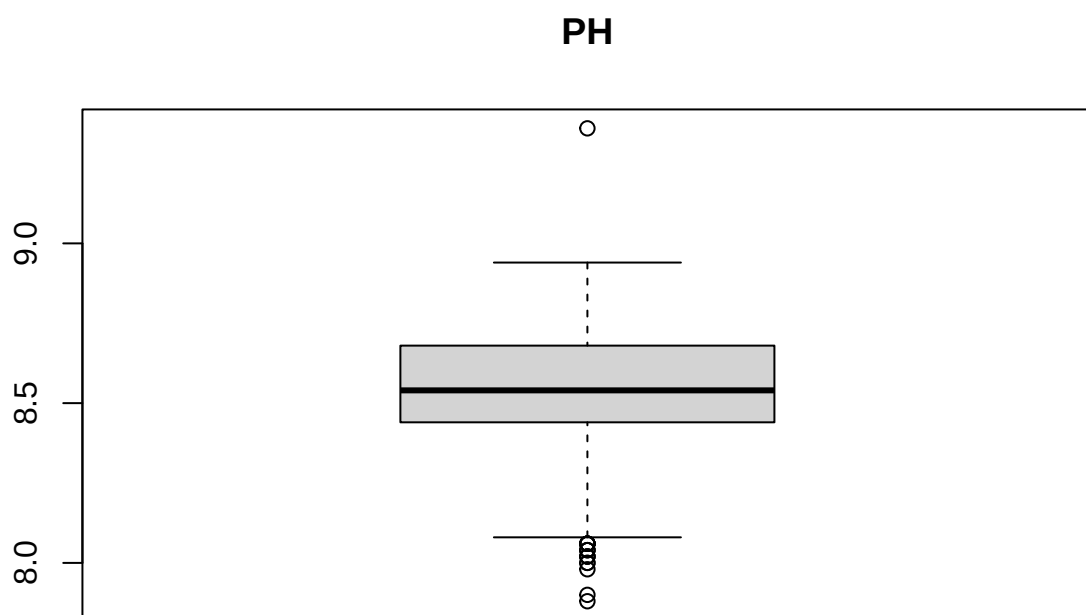


Usage.cont

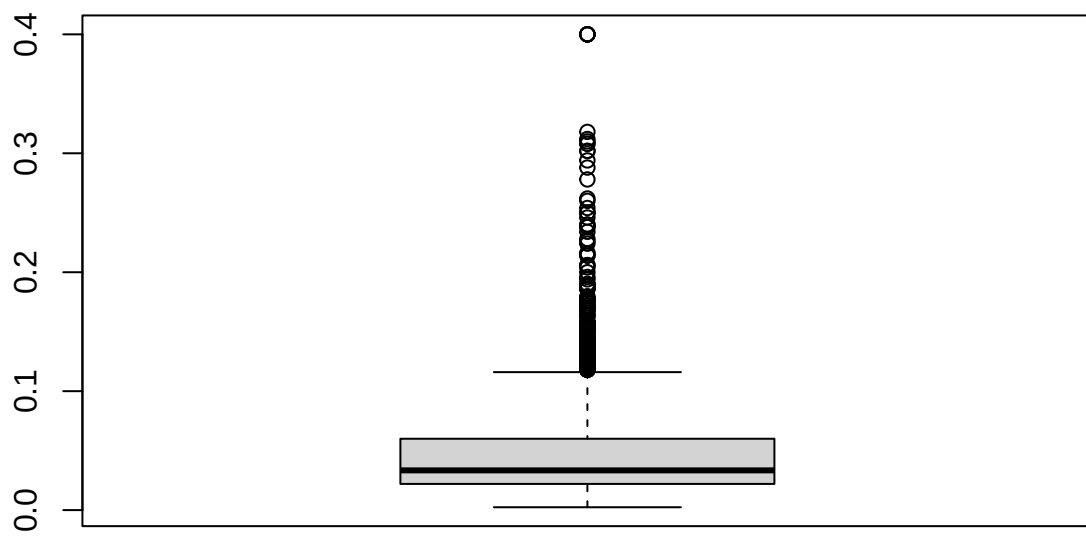


Carb.Flow

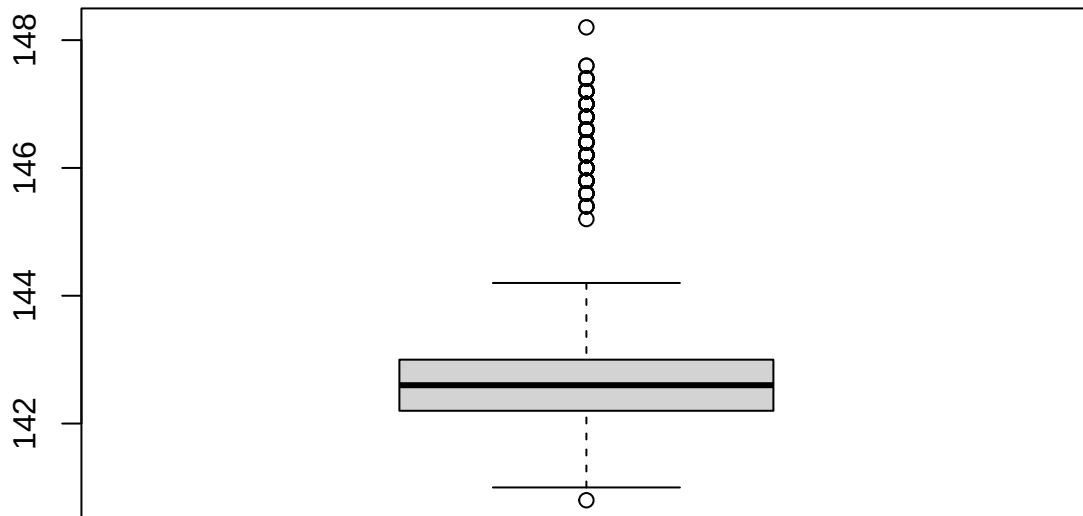




Oxygen.Filler



Air.Pressurer



Calculating the number of outliers in each column based on the IQR method

```
outliers <- sapply(training_data_new %>% select(-Brand.Code), function(x) {
  Q1 <- quantile(x, 0.25, na.rm=TRUE)
  Q3 <- quantile(x, 0.75, na.rm=TRUE)
  IQR <- Q3 - Q1
  sum(x < Q1 - 1.5*IQR | x > Q3 + 1.5*IQR, na.rm=TRUE)
})
outliers
```

##	Carb.Volume	Fill.Ounces	PC.Volume	Carb.Pressure	Carb.Temp
##	5	55	86	14	36
##	PSC	PSC.Fill	PSC.CO2	Mnf.Flow	Carb.Pressure1
##	52	54	77	0	24
##	Fill.Pressure	Hyd.Pressure2	Filler.Level	Filler.Speed	Temperature
##	79	0	9	439	123
##	Usage.cont	Carb.Flow	PH	Oxygen.Filler	Air.Pressurer
##	0	0	18	189	220

Cross-checking the graphs with the data set, we could remove the one Filler.Level value that appears on the top of the box plot, 161.2 when the next-highest is 154. pH has one comparatively high value (9.36) that's nowhere close to the next-highest value. We can remove that row - it's missing 4 other variables. PH at that level isn't impossible in beverage manufacturing, but with 4 missing predictor values in the same row and the value isolated far from the rest of the distribution, it reads more like a data entry issue than a legitimate measurement to learn from. Other outlier values are one of several similar values, and could be categorized

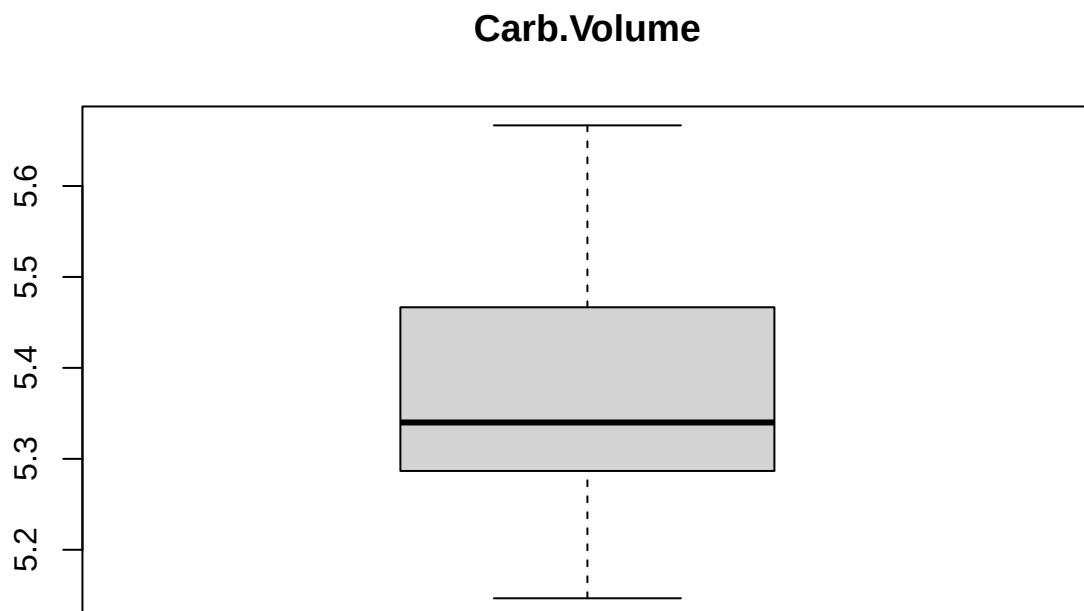
that way due to the shape of the distribution and wide range of values. For example, it looks like there's one particularly high value for Oxygen.Filler, but when we check there are 5 cases with the same value.

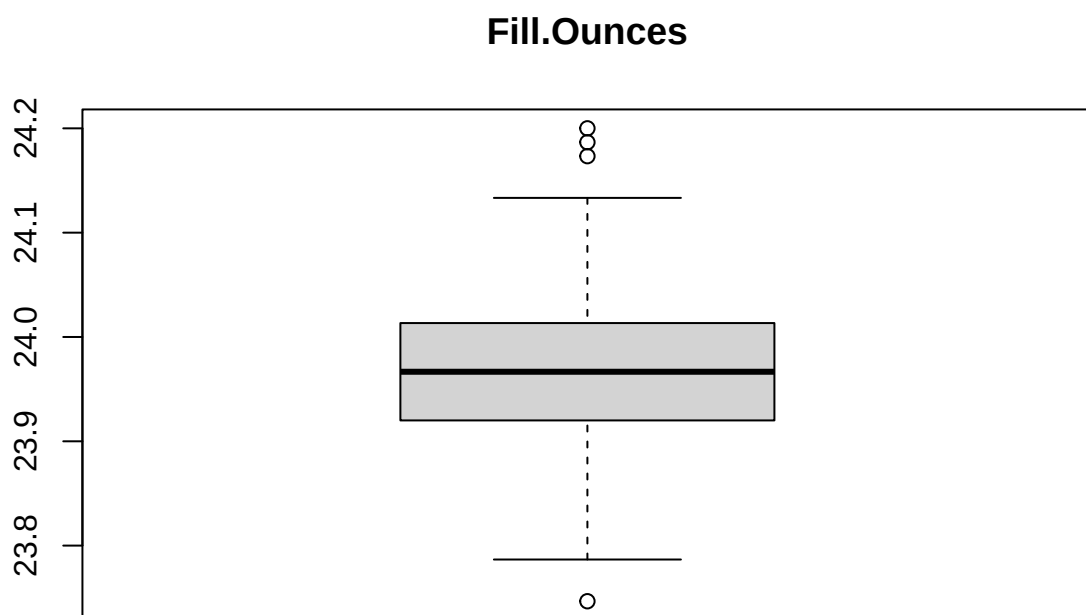
```
#removing the one outliers
training_data_new <- training_data_new %>%
  filter(PH < 9 | is.na(PH)) %>%
  filter(Filler.Level < 160 | is.na(Filler.Level))

#removing pH NAs because we shouldn't impute the response variable
training_data_no_na <- training_data_new %>%
  filter(!is.na(PH))
```

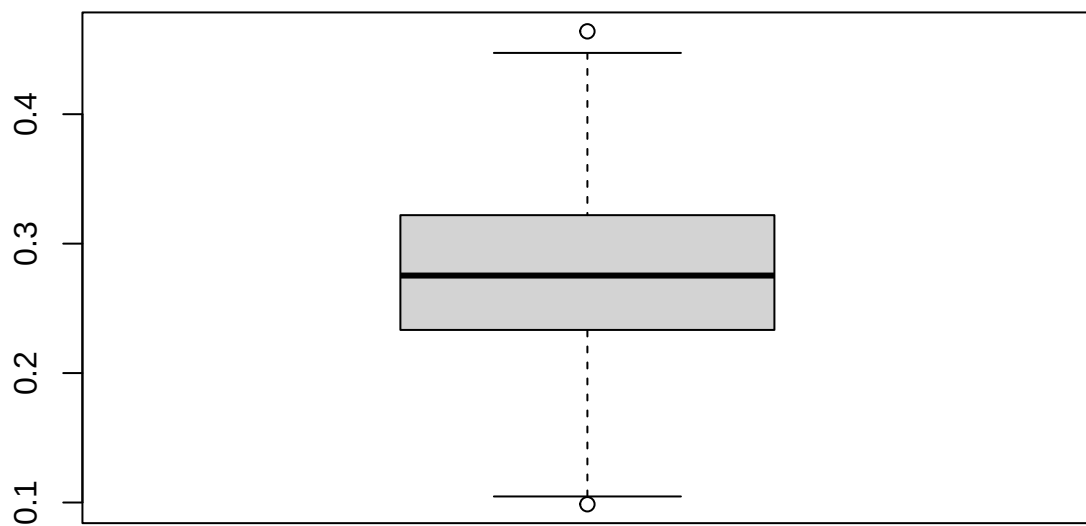
1.3.5 Looking at outliers in the test data set

```
#leaving out the categorical and blank response variable
for (col in names(test_data_new %>% select(-Brand.Code, -PH))) {
  boxplot(test_data_new[[col]], main=col)
}
```

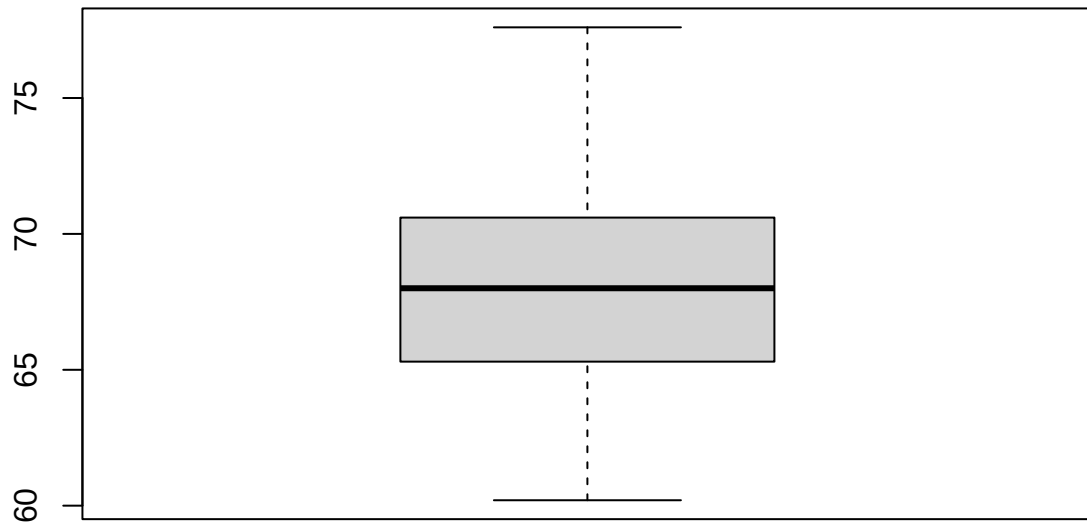




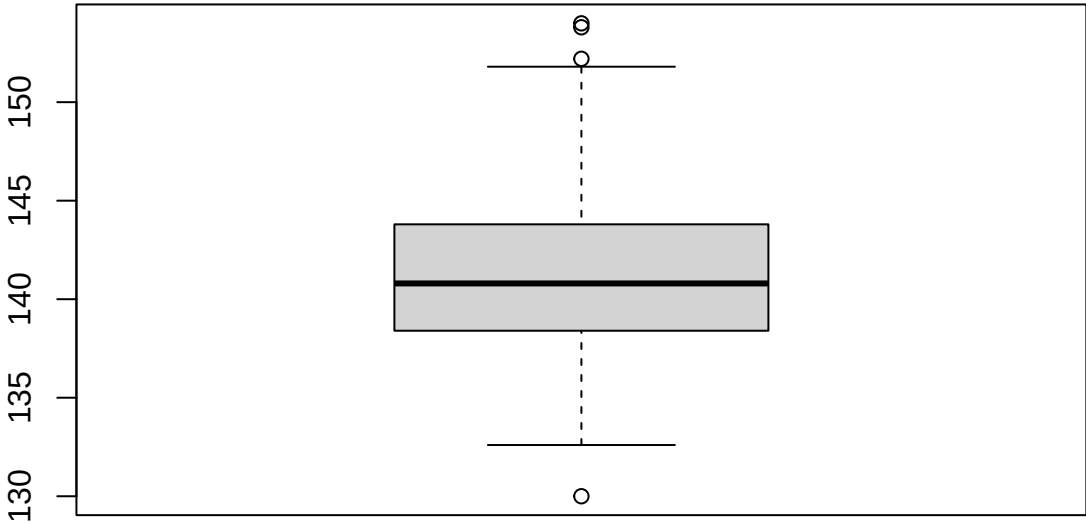
PC.Volume

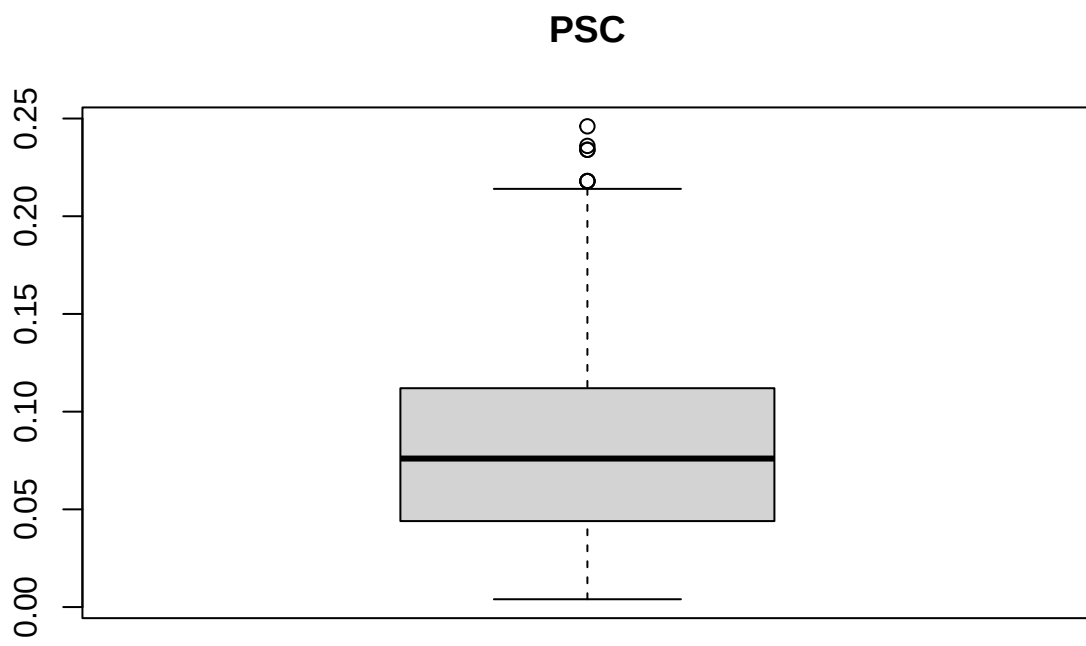


Carb.Pressure

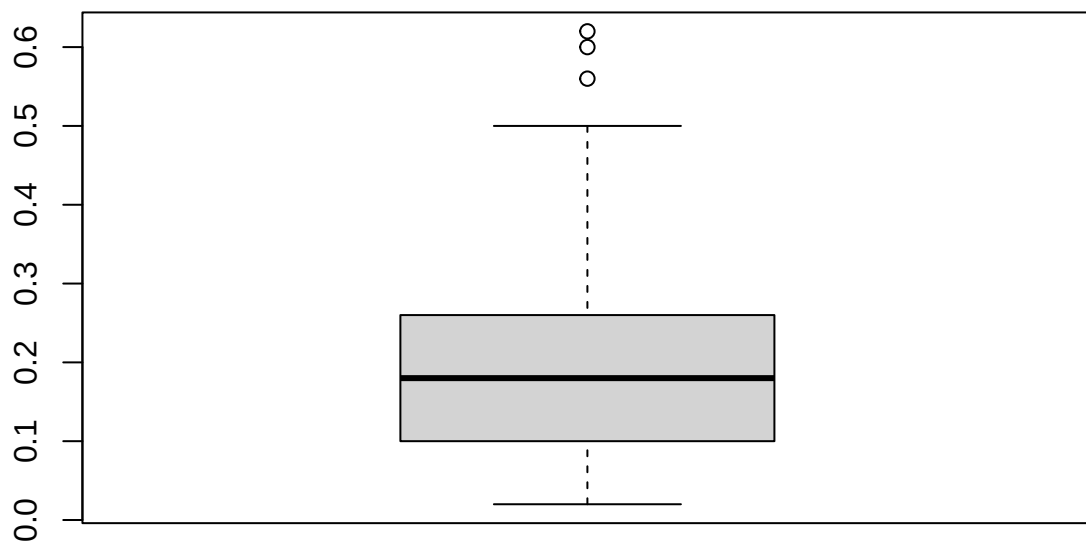


Carb.Temp

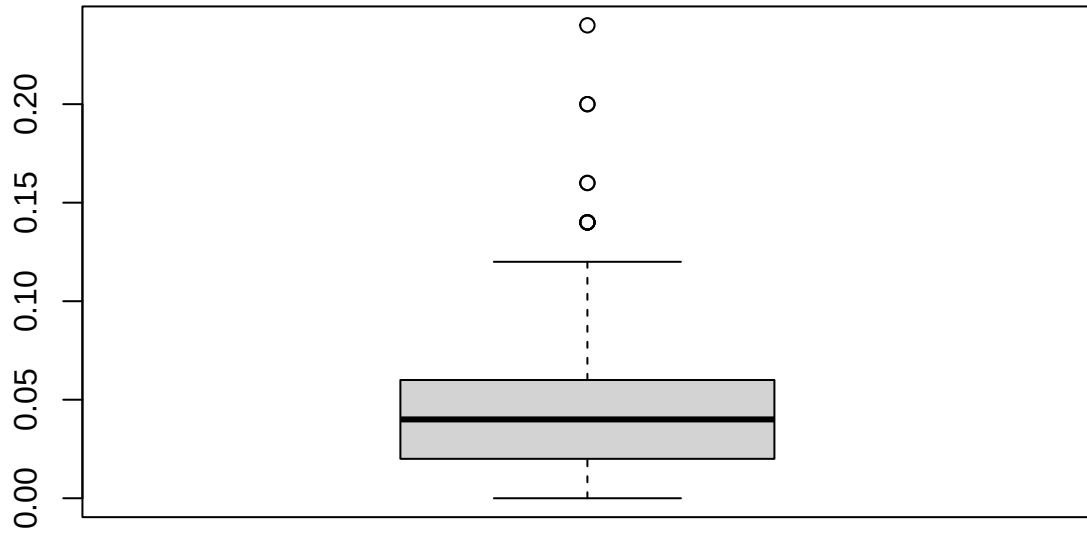




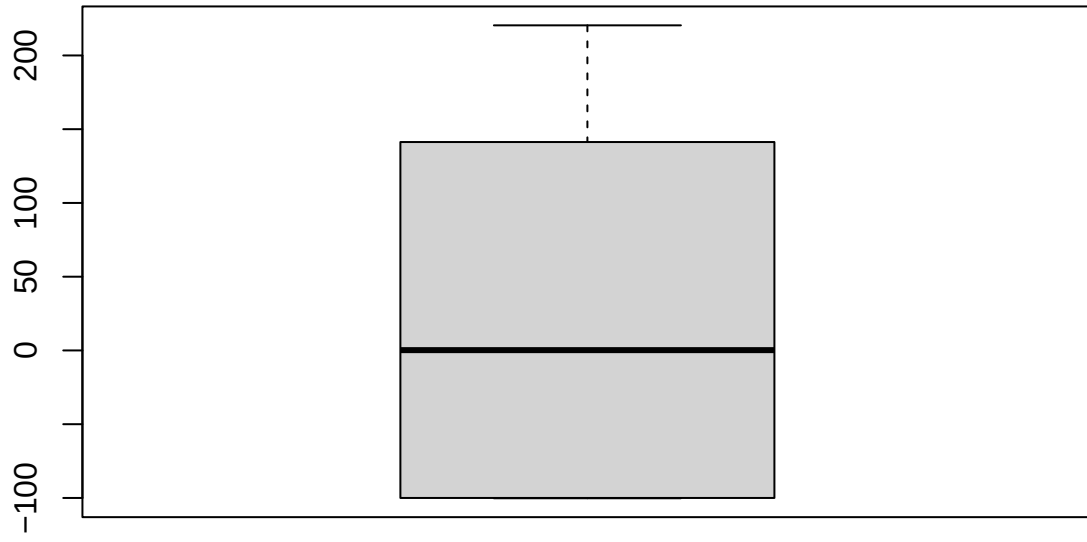
PSC.Fill



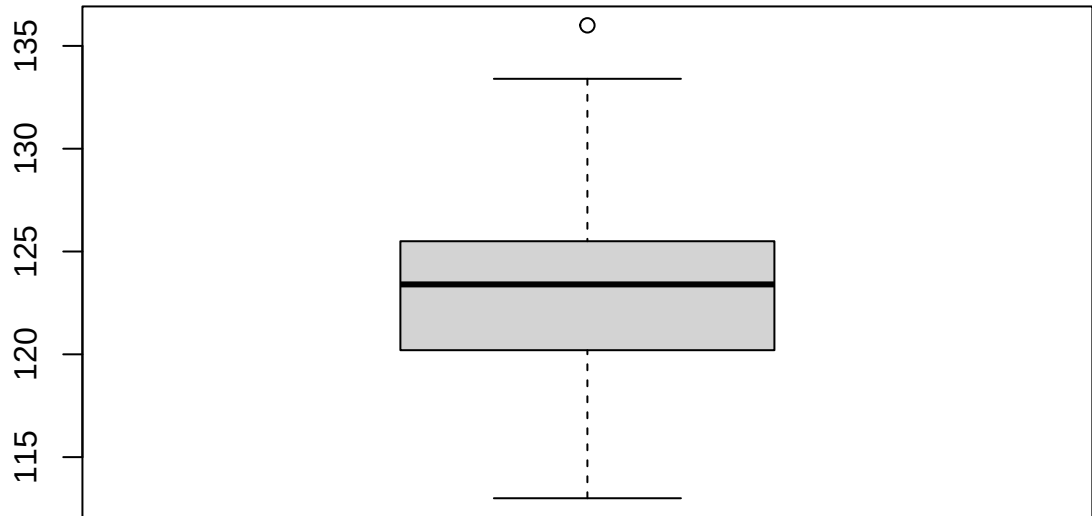
PSC.CO2



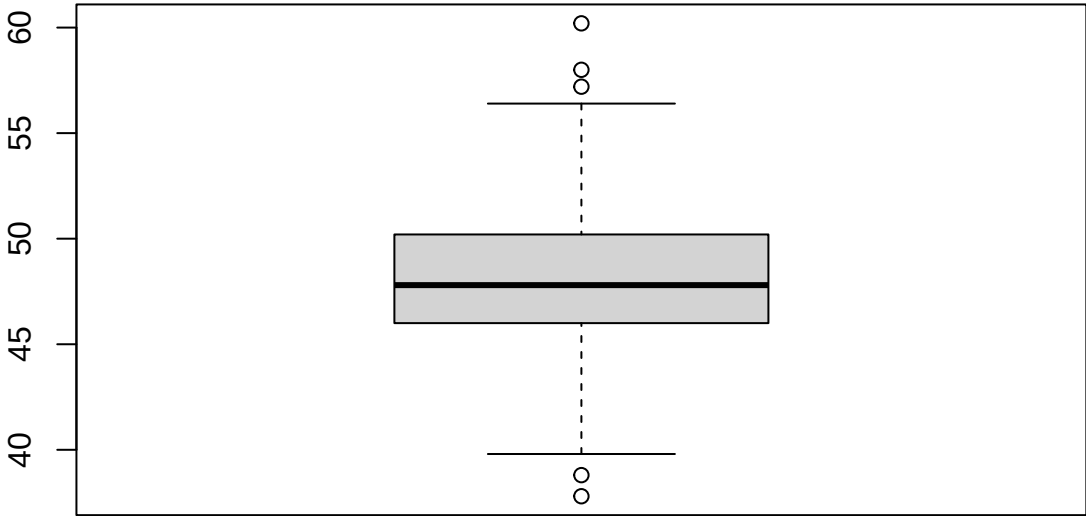
Mnf.Flow



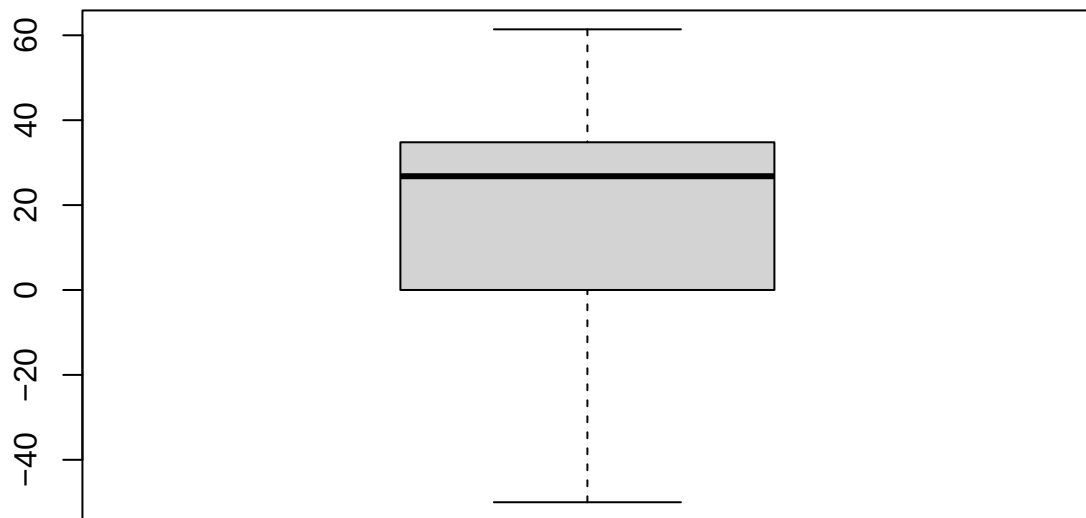
Carb.Pressure1



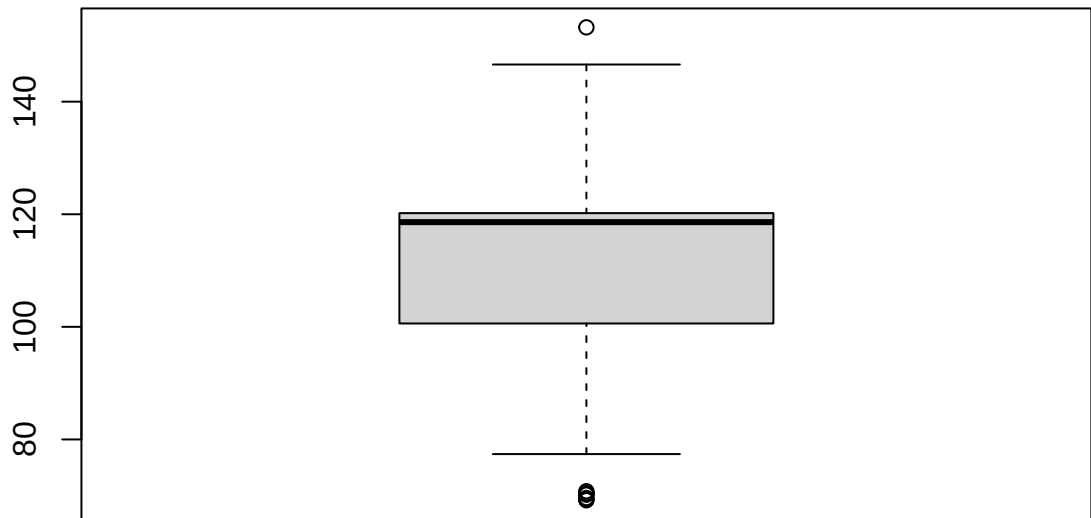
Fill.Pressure

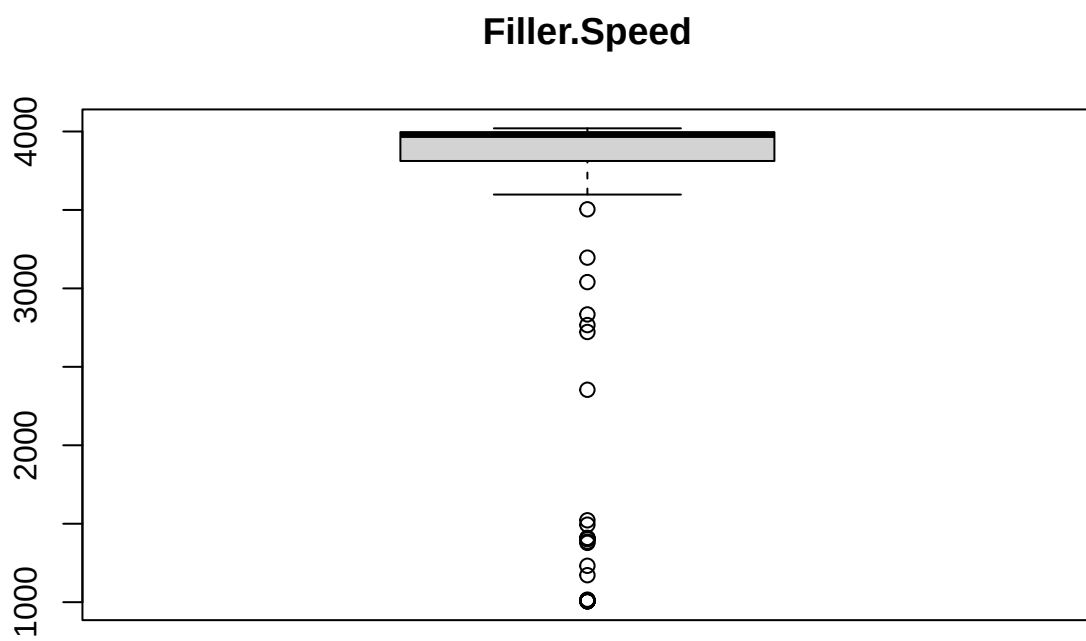


Hyd.Pressure2

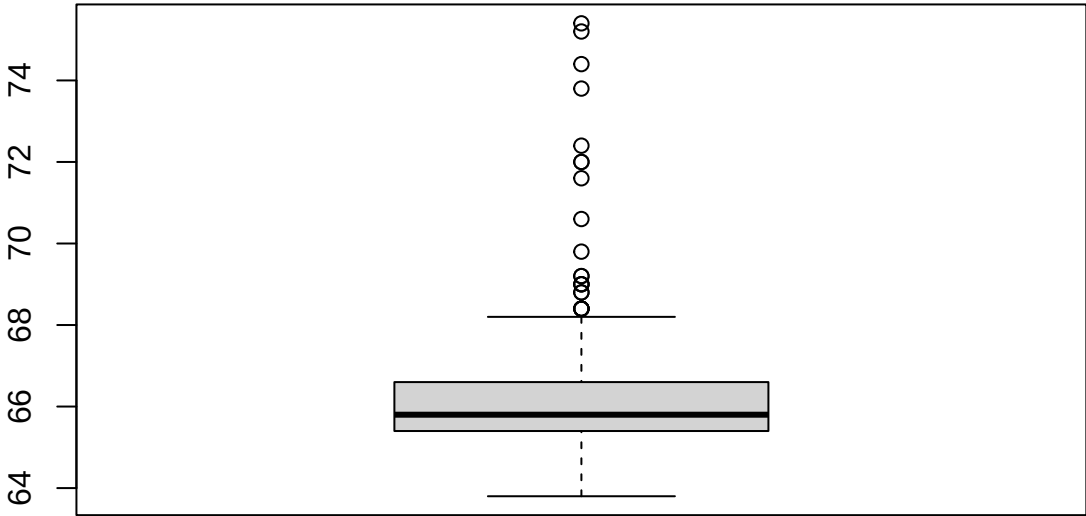


Filler.Level

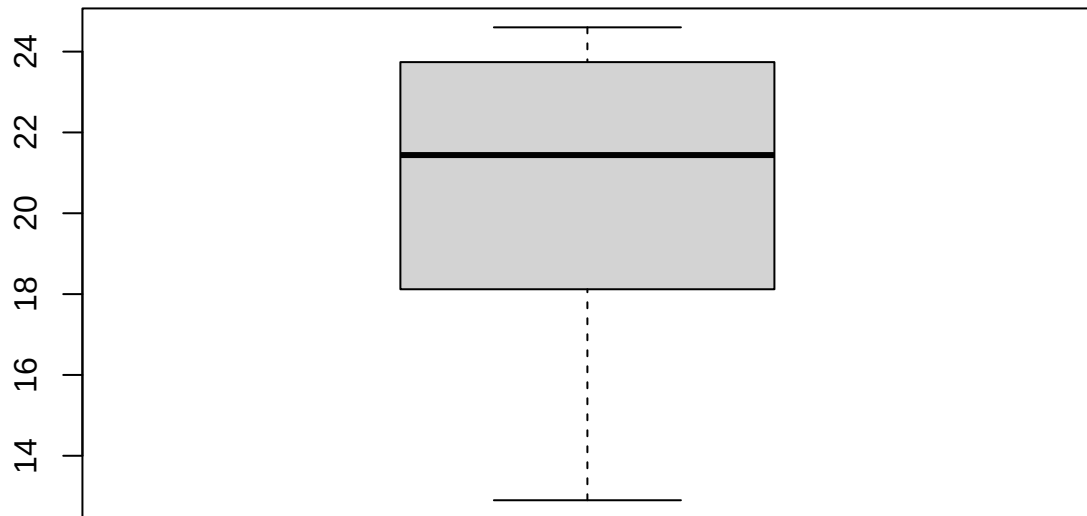


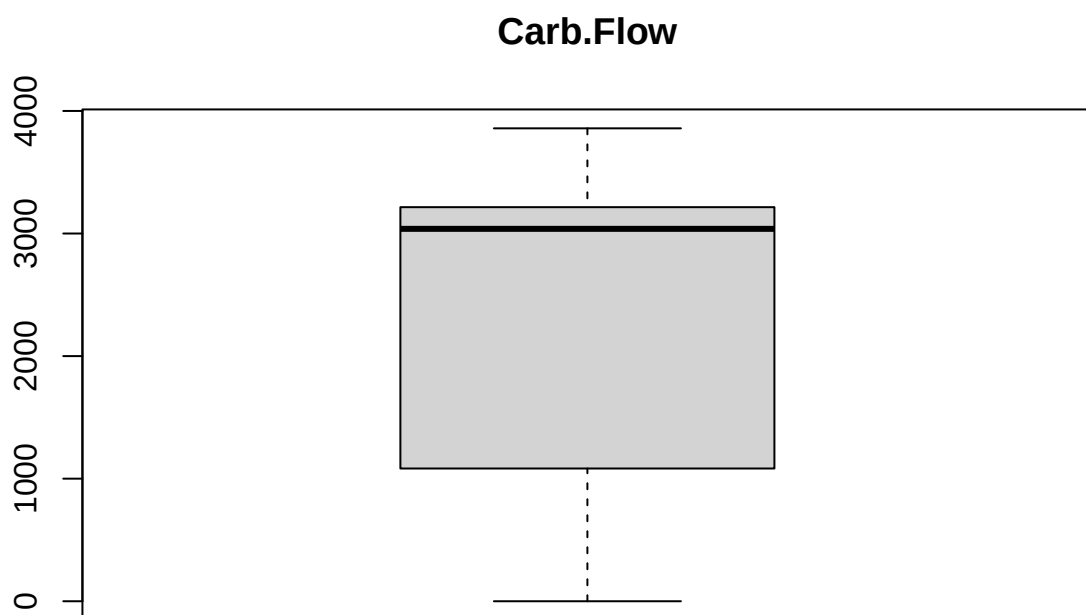


Temperature

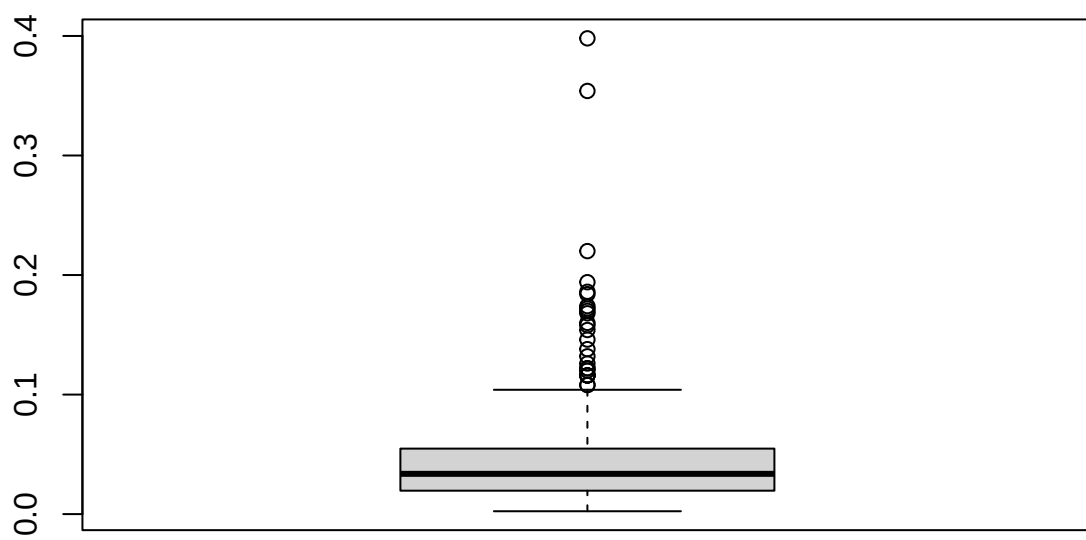


Usage.cont

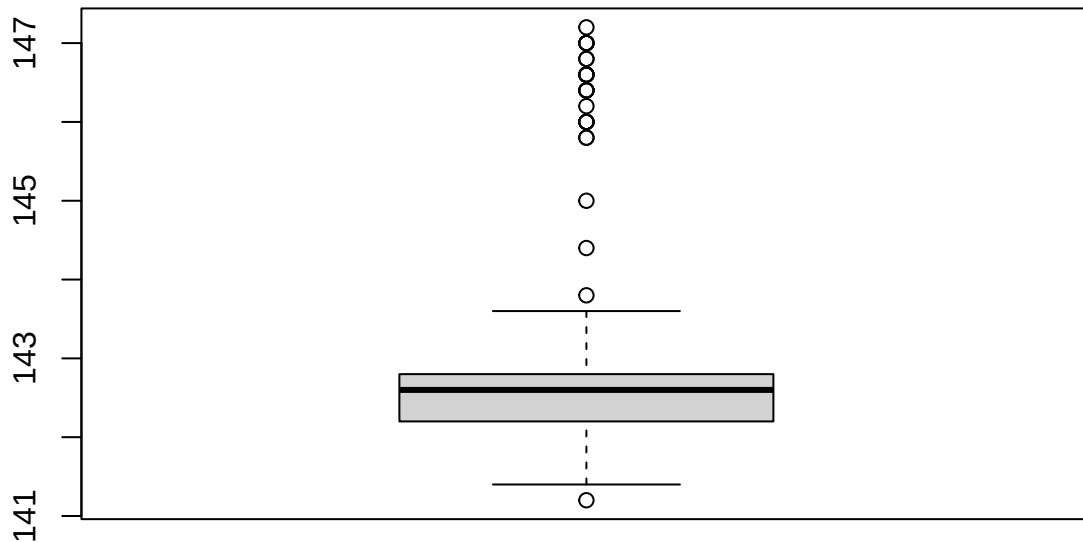




Oxygen.Filler



Air.Pressurer



```
test_data_new %>% count(Brand.Code)
```

```
##   Brand.Code    n
## 1         A    35
## 2         B   129
## 3         C    31
## 4         D    64
## 5        <NA>     8
```

Oxygen filler shows up again with a few outliers, which just seems to be the case for this variable; the outliers in filler speed are all clustered; filler level has outliers on both ends, as does fill pressure. Carb.pressure1 has one high value, PSC.CO2 has a few higher values designated outliers due to the clustering on the lower end, same is true for PSC.Fill, and the rest are similar.

We won't remove any of these because they don't look like errors, and in some cases represent a different, smaller cluster of values.

1.3.6 Keeping a wider feature set for tree-based models

The correlation prune above takes us to 20 predictors, which is the right call for linear models, PLS, penalized regression, and neural networks. Kuhn and Johnson notes that neural network models can be hurt by high correlation among the predictors. Tree-based models tolerate collinearity fine, as discussed in the regression trees coverage in Kuhn and Johnson.

So we keep both:

- `training_data_new` / `test_data_new`: 20 predictors after NZV removal and $r > 0.90$ correlation prune. Linear models, PLS, ridge / lasso / elastic net, and neural networks use this.
- `training_data_wide` / `test_data_wide`: 31 predictors after just NZV removal. Random forest, gradient boosting, Cubist use this. The correlated-but-distinct sensor readings may each carry small bits of information a tree can exploit independently.

```
# wide feature set: post-NZV only, same outlier filters as the narrow set
training_data_wide <- training_data %>%
  select(-Hyd.Pressure1) %>%
  filter(PH < 9 | is.na(PH)) %>%
  filter(Filler.Level < 160 | is.na(Filler.Level))

training_data_wide_no_na <- training_data_wide %>%
  filter(!is.na(PH))

# same column drop on the evaluation set
test_data_wide <- test_data %>%
  select(-Hyd.Pressure1)
```

1.3.7 Recap

- Dropped `Hyd.Pressure1` for near-zero variance
- Dropped 11 collinear variables (cutoff $r > 0.90$), keeping one representative per correlated group; manually chose `MFR` over `Filler.Speed` since `MFR` had more missingness
- Outlier check via IQR + histograms + boxplots: removed two rows only (one with `PH = 9.36`, one with `Filler.Level = 161.2`). Everything else looked like a legitimate value or part of a cluster
- Removed rows with missing `PH` since we can't impute the response
- Narrow training set has 21 columns (20 predictors + `PH`), still pending imputation
- Also derived a wider feature set (`training_data_wide_no_na`, 31 predictors) by skipping the correlation prune, for tree-based models that handle collinearity natively

1.4 Dealing with missing values

Recap located at the end of the section. Skip there to jump past the code/workflow.

```
sum(is.na(training_data_no_na))
```

```
## [1] 515
```

```
sort(colSums(is.na(training_data_no_na)), decreasing = TRUE)
```

```
##      Brand.Code  Filler.Speed    PC.Volume    PSC.CO2    Fill.Ounces
##           120         52         39         39         38
##           PSC Carb.Pressure1 Carb.Pressure    Carb.Temp    PSC.Fill
##           33         32         27         26         23
##  Fill.Pressure Hyd.Pressure2  Filler.Level  Temperature  Oxygen.Filler
##           17         15         15         11         11
##   Carb.Volume    Usage.cont    Carb.Flow    Mnf.Flow    PH
##           10         5         2         0         0
## Air.Pressurer
##           0
```

```
#checking the number of complete cases
table(rowSums(is.na(training_data_no_na)))
```

```
##
##      0      1      2      3      4      5      6
## 2169  309   64   18    2    2    1
```

516 missing values across 2566 cases and 21 variables - 516/53886 is about .9% overall missingness, which is very low. Notably, brand code is missing in 120 cases.

2169 cases out of 2566 are complete, or about 85%. If we dropped all NAs, we would be dropping 15% of the data. Instead, we will use the VIM package for simple KNN imputation. This package just fills in the nearest neighbor—it doesn't change the other variables. VIM can handle categorical variables, like brand code. We'll use K=5.

```
training_data_imputed <- kNN(training_data_no_na, k = 5)
```

```
#making sure it worked
sum(is.na(training_data_imputed))
```

```
## [1] 0
```

1.4.0.1 Imputing values for the evaluation set First, let's check how many missing values there are after we've removed columns in the evaluation set.

```
table(rowSums(is.na(test_data_new)))
```

```
##
##      1      2      3      4      5      9
## 225  33    6    1    1    1
```

```
colSums(is.na(test_data_new))
```

```
##      Brand.Code      Carb.Volume      Fill.Ounces      PC.Volume      Carb.Pressure
##              8              1              6              4              0
##      Carb.Temp      PSC      PSC.Fill      PSC.CO2      Mnf.Flow
##              1              5              3              5              0
## Carb.Pressure1 Fill.Pressure Hyd.Pressure2 Filler.Level Filler.Speed
##              4              2              1              2              10
##      Temperature      Usage.cont      Carb.Flow      PH      Oxygen.Filler
##              2              2              0      267              3
##      Air.Pressurer
##              1
```

```
#total NAs (subtracting 267 for the missing pH values)
sum(is.na(test_data_new)) - 267
```

```
## [1] 60
```

Now there are only 60 missing values total. 42 rows have missing values. One row has 8 missing values (the one that had 14 before).

Imputing a row that sparse means the filled-in predictor values will be noisy, but the deliverable expects a prediction for every row in the evaluation set. We're keeping the row in and flagging the prediction as low-confidence rather than dropping it.

```
#keeping all 267 rows; the sparse row gets imputed but its prediction is flagged downstream
count(test_data_new)
```

```
##      n
## 1 267
```

Imputing with the same method we used for the training data:

```
#list all columns except PH
cols_to_impute <- setdiff(colnames(test_data_new), "PH")

test_data_imputed <- kNN(test_data_new, variable = cols_to_impute, k = 5)

#we should get 267 cases
sum(is.na(test_data_imputed))
```

```
## [1] 267
```

1.4.1 Recap

- Training set started with 516 missing cells across 2566 rows x 21 columns (~0.9% missingness). Imputed all gaps with VIM `kNN(k = 5)` -> `training_data_imputed`
- Evaluation set: kept all 267 rows including one with 8 predictor NAs out of 20 (originally 14 before column drops); imputed predictor columns only with VIM `kNN(k = 5)` (leaving PH blank since that's what we predict) -> `test_data_imputed`. The sparse row gets imputed too but will be flagged as low-confidence in the final prediction output.
- Both sets are now fully imputed (apart from PH in the eval set) and ready for modeling

1.5 Break into training and test data

Recap located at the end of the section. Skip there to jump past the code/workflow.

Breaking the training data set into an 80%/20% training and test set:

```
set.seed(624)

train_index <- sample(nrow(training_data_imputed),
                     0.8 * nrow(training_data_imputed))

#create a training set
train_y <- training_data_imputed$PH[train_index]
train_x <- training_data_imputed[train_index, ] %>% dplyr::select(-PH)

#test set
test_y <- training_data_imputed$PH[-train_index]
test_x <- training_data_imputed[-train_index, ] %>% dplyr::select(-PH)
```

1.5.1 Centering and scaling

I did not scale anything individually, every package should have an option to center/scale automatically, and use a Box-Cox for the data with less normal distributions.

1.5.2 Recap

- Split `training_data_imputed` 80/20 via `sample()` with `set.seed(624)`, producing `train_x` / `train_y` (80%) and `test_x` / `test_y` (20%)
- Centering, scaling, and Box-Cox transformations left to per-model `preProc` options rather than baked in here

1.6 Modeling approach

Recap located at the end of the section. Skip there to jump past the workflow.

Before fitting models, a few setup decisions worth flagging up front.

caret rather than recipes. The textbook for this part of the course (Kuhn and Johnson) uses `caret::preProcess` for imputation, transformations, and scaling inside the resampling loop. We're sticking with that for the sake of sticking with the class standard and what we've been learning how to use effectively. However we did stumble upon the **recipes** made by the same author later. Potentially worth looking into in the future.

Per-model feature sets. As covered in the data cleaning section, linear-family and neural-network models use the correlation-pruned 20-predictor set (`training_data_no_na`). Tree-based and rule-based models use the wider 31-predictor set (`training_data_wide_no_na`). The reasoning is in the cleaning section's notes.

Imputation inside resampling. The earlier `kNN(k = 5)` chunk produced `training_data_imputed` as a one-time inspection step, useful for seeing what imputation does to the data and for the quick 80/20 sanity-check split. For the actual model fitting, however, we start from `training_data_no_na` (the pre-imputation intermediate) and let `caret` handle imputation inside each CV fold via `preProcess(method = c("knnImpute", "BoxCox", "center", "scale"))` paired with `trainControl(method = "repeatedcv", number = 10, repeats = 5)`. That way the imputation parameters are estimated per fold rather than once on the full training set, which avoids the small bias from imputing before the split.

1.6.1 Recap

- Linear-family + NN models get the 20-predictor narrow set; tree-based + rule-based models get the 31-predictor wide set.
- The pre-imputed `training_data_imputed` is an inspection / sanity-check artifact for imputation. The actual model fitting starts from the pre-imputation data (`training_data_no_na`) and lets `caret` impute inside each CV fold for honest performance metrics.